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Kim

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(54) **SEMICONDUCTOR PACKAGE**

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H01L 23/00 (2006.01)

H01L 23/498 (2006.01)

(52) **U.S. Cl.**

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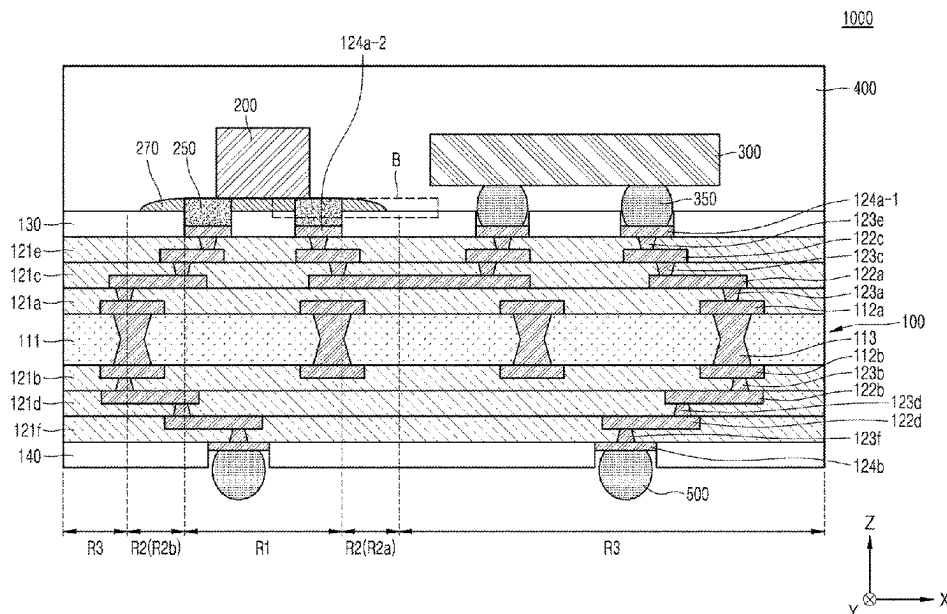
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(57) **ABSTRACT**

A semiconductor package includes a substrate having a passive element region, a peripheral region adjacent to the passive element region, and a remaining region, a first passive element on an upper surface of the passive element region, a first semiconductor chip on an upper surface of the remaining region, and a sealing portion covering the substrate, the first passive element, and the first semiconductor chip, wherein the peripheral region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side, and wherein a roughness of an upper surface of at least one of the first to fourth sub-regions is greater than a roughness of the upper surface of the remaining region.

20 Claims, 24 Drawing Sheets



(52) **U.S. Cl.**

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(2013.01); *H01L 24/48* (2013.01); *H01L*
2224/16227 (2013.01); *H01L 2224/16237*
(2013.01); *H01L 2224/48228* (2013.01); *H01L*
2924/1431 (2013.01); *H01L 2924/1436*
(2013.01); *H01L 2924/1438* (2013.01)

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CPC *H01L 2924/1431*; *H01L 2924/1436*; *H01L*
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H01L 24/48; *H01L 24/13*; *H01L 23/3128*;
H01L 23/49816; *H01L 23/49822*; *H01L*
23/49838; *H01L 23/50*; *H01L 25/16*;
H01L 25/0655; *H01L 25/0657*

See application file for complete search history.

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FIG. 1A

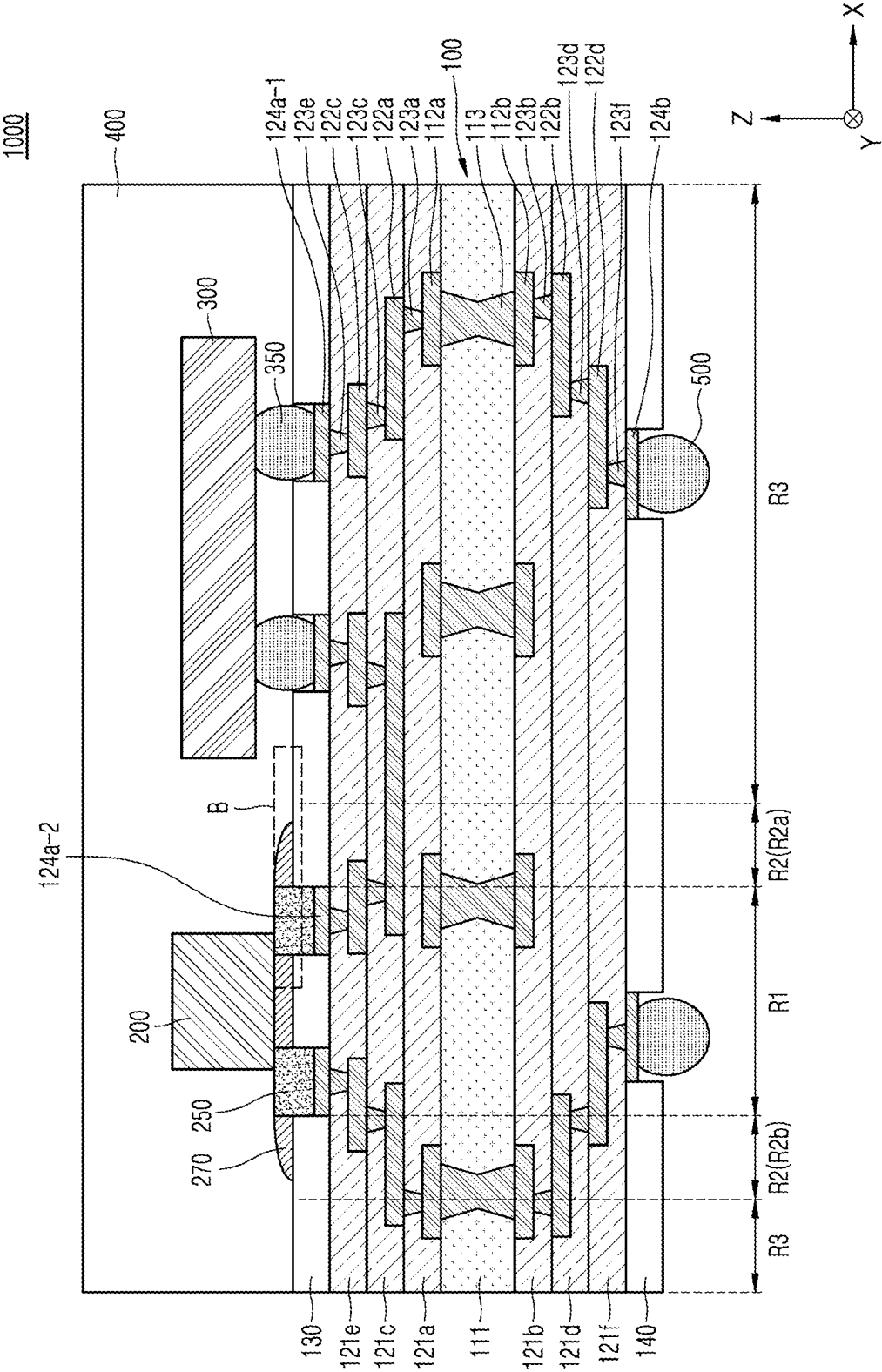


FIG. 1B

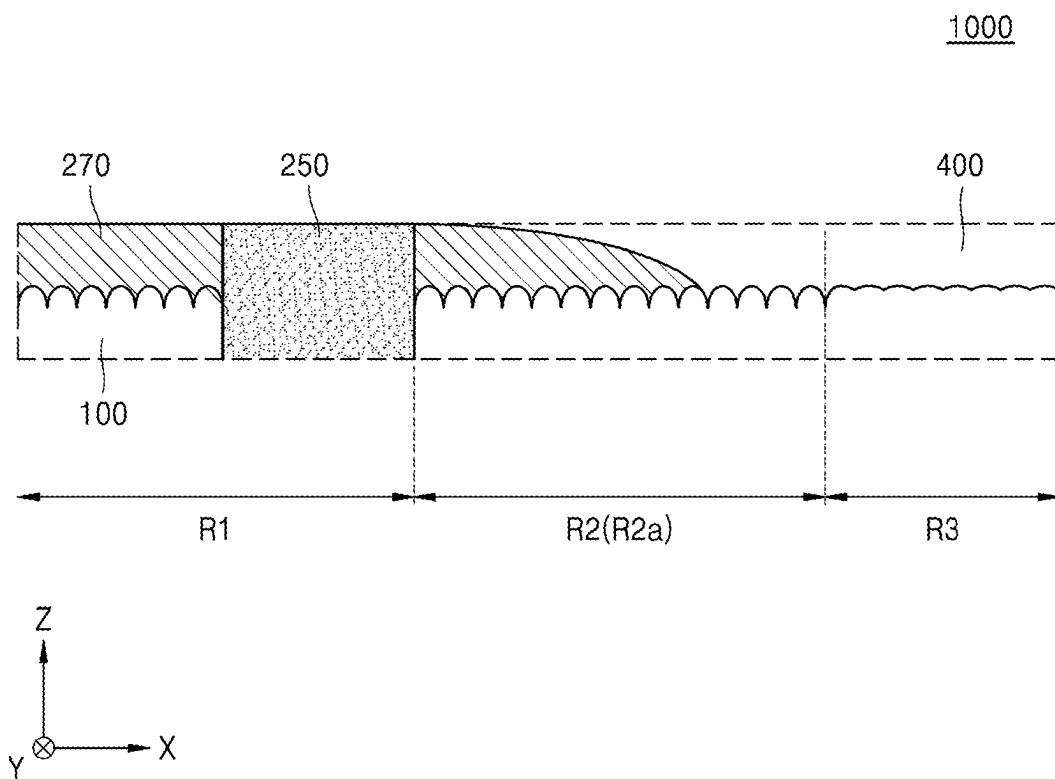


FIG. 1C

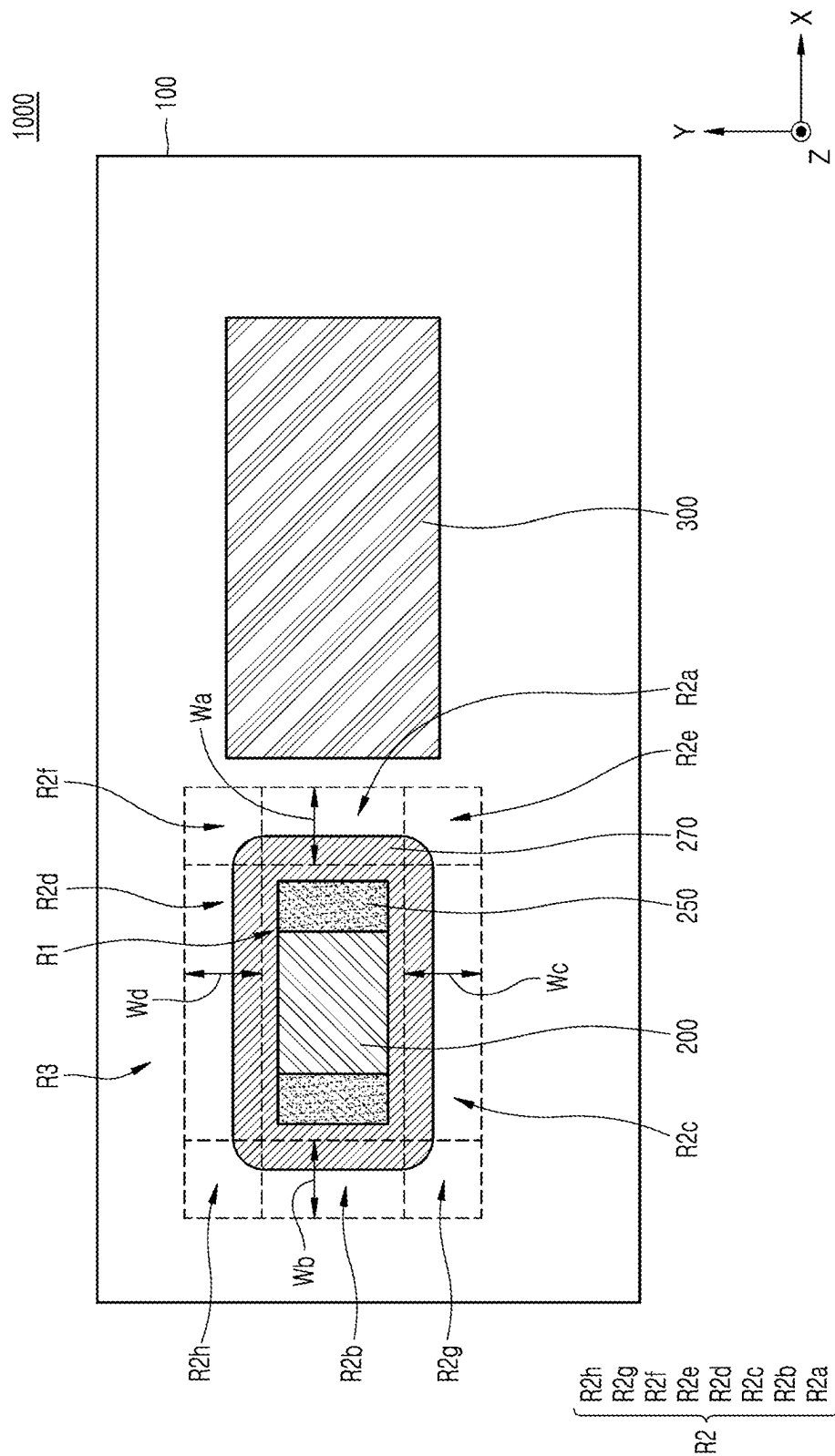


FIG. 2

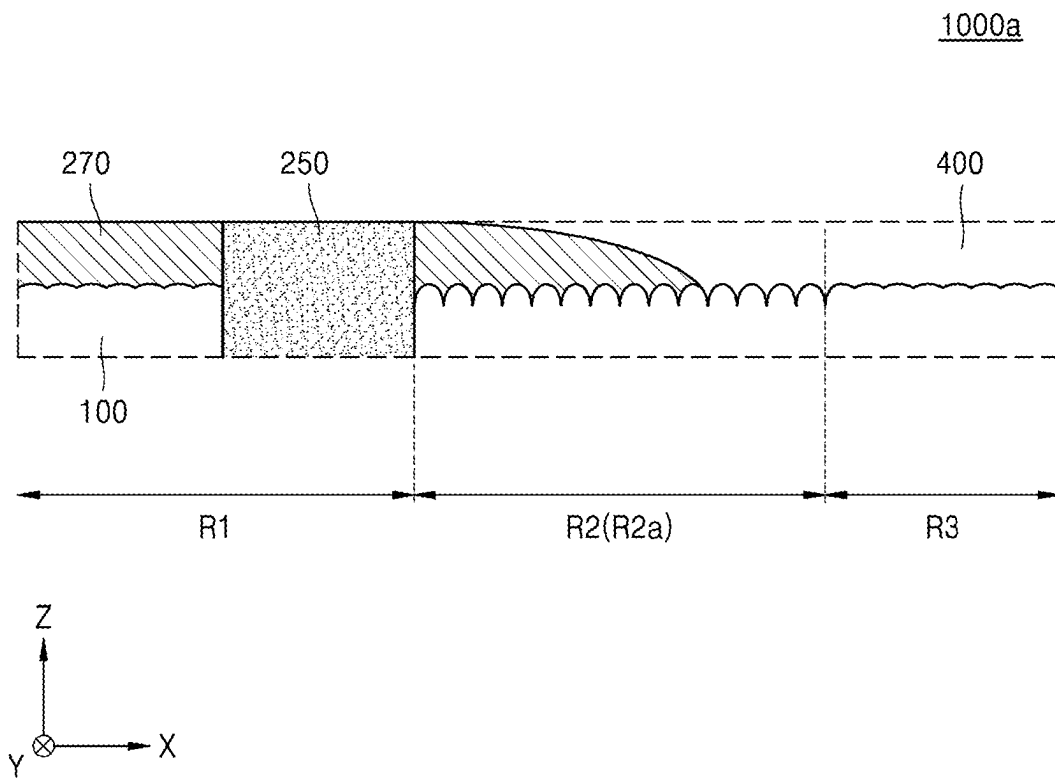


FIG. 3

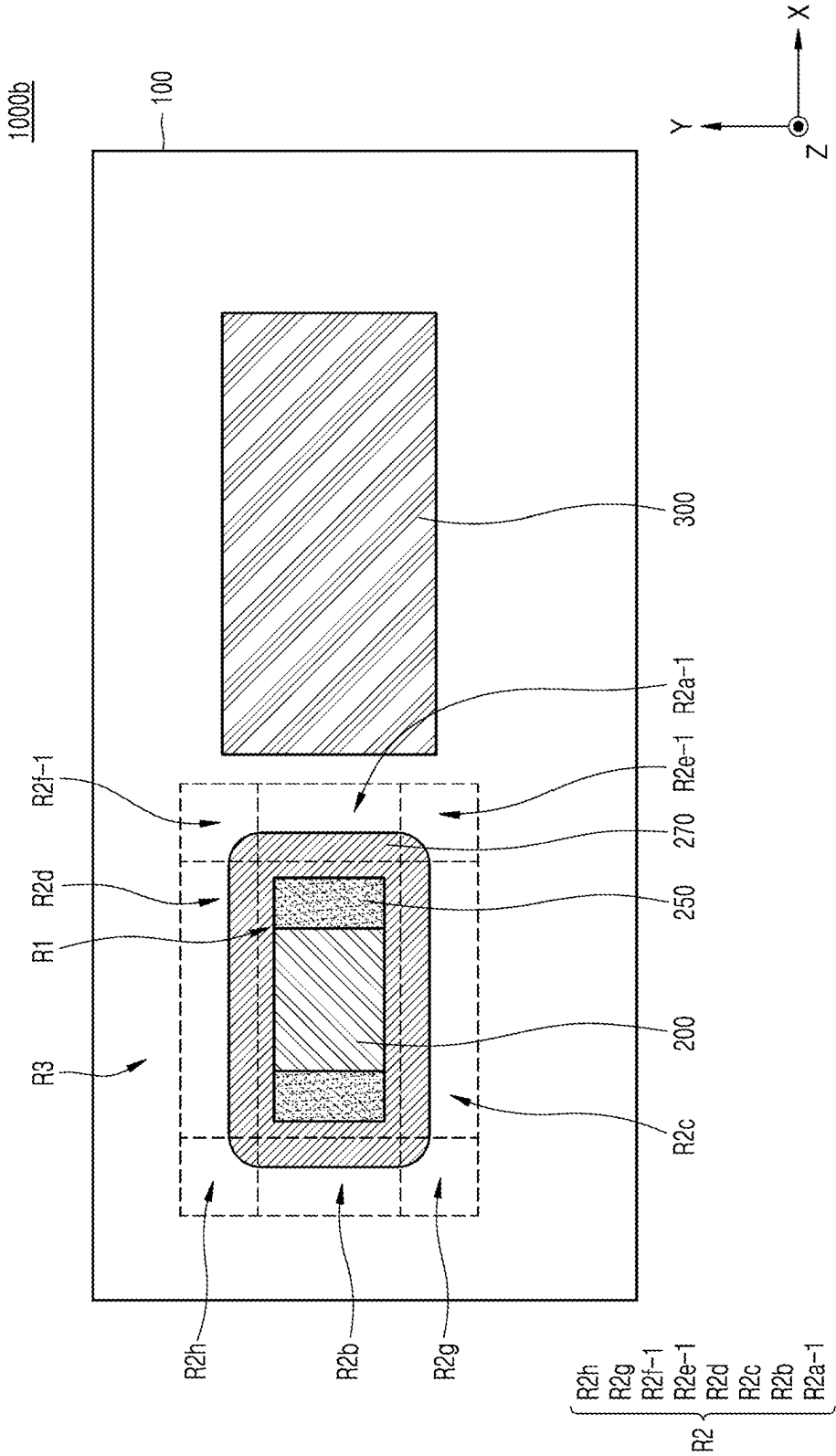


FIG. 4

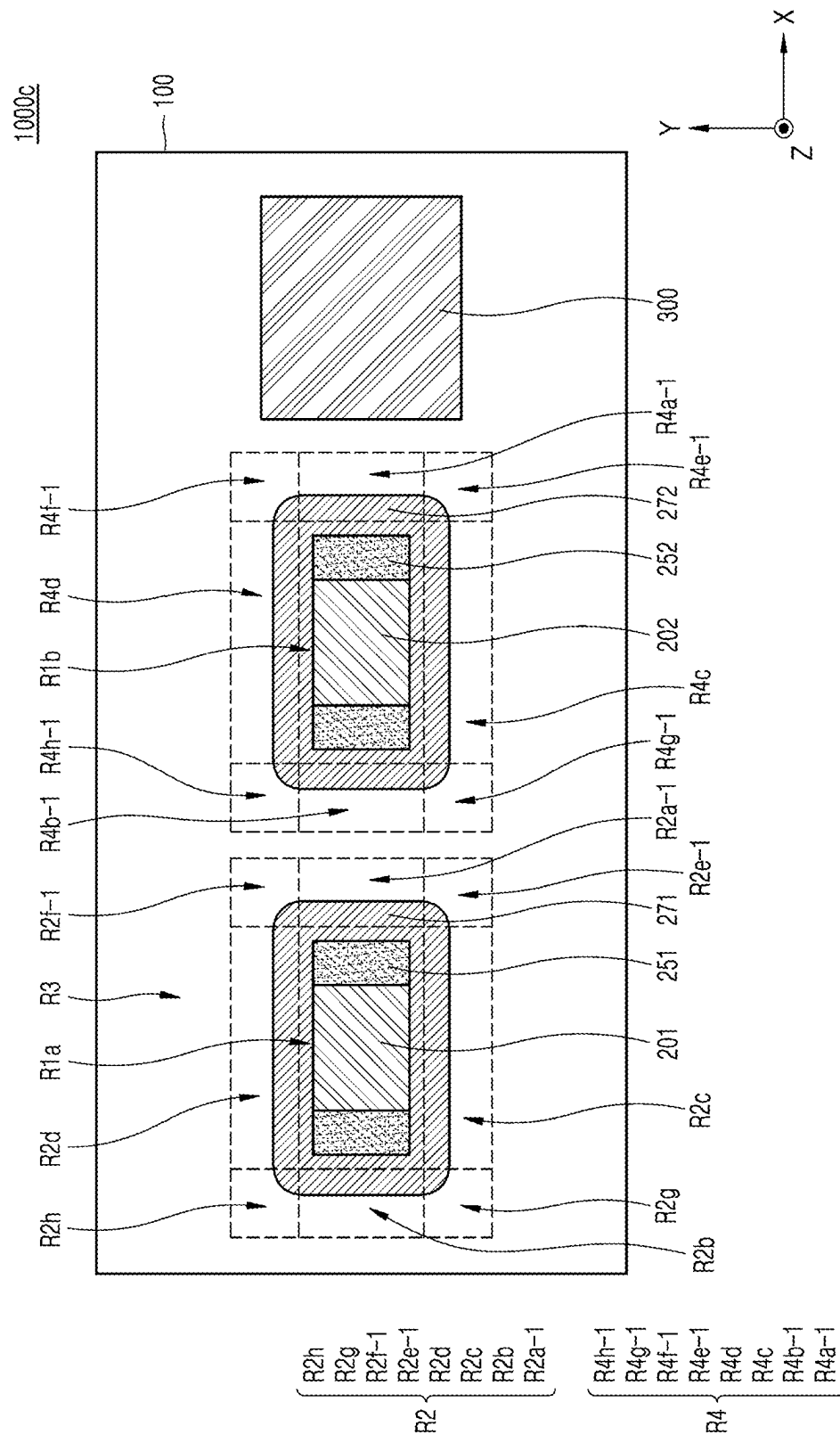


FIG. 5

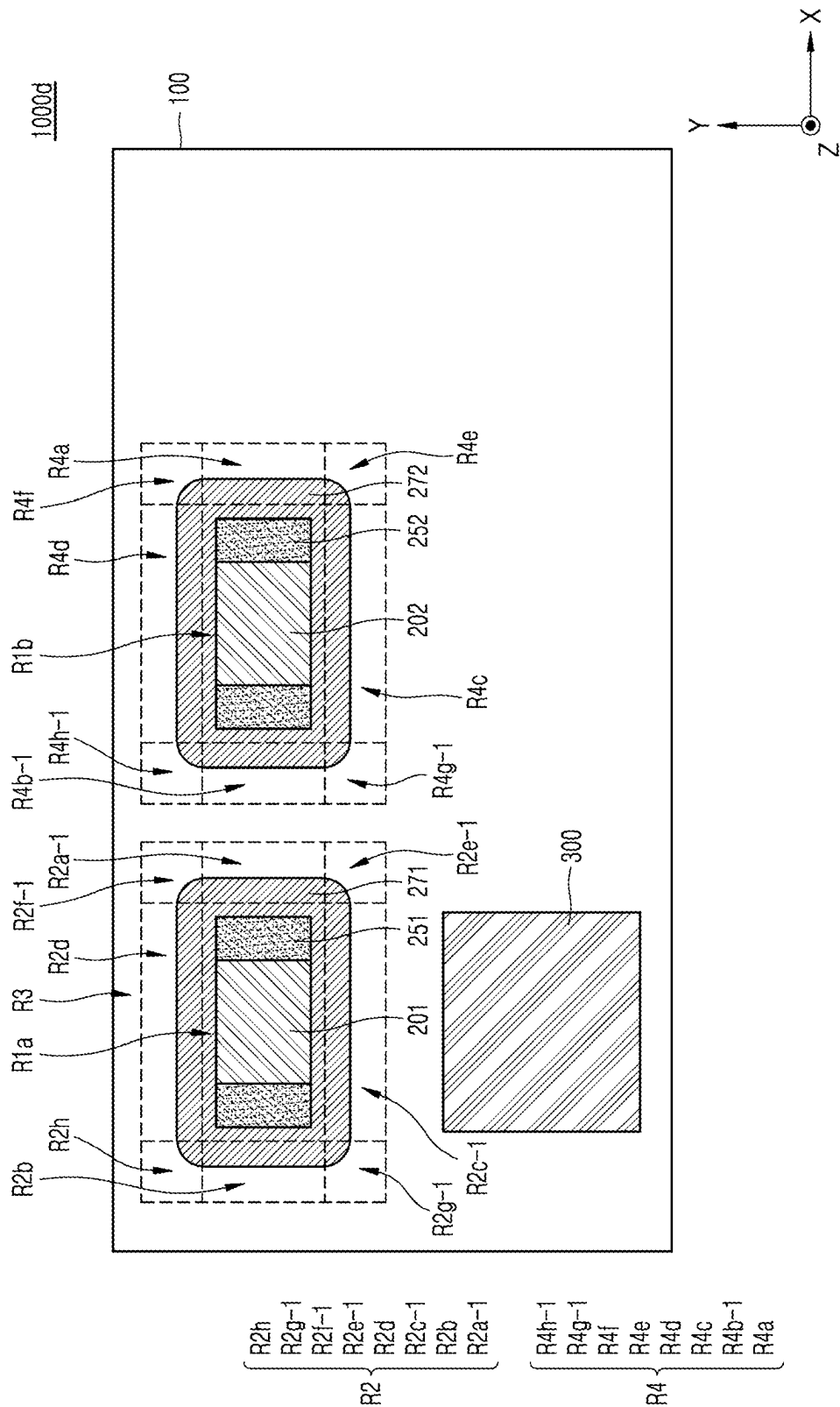


FIG. 6

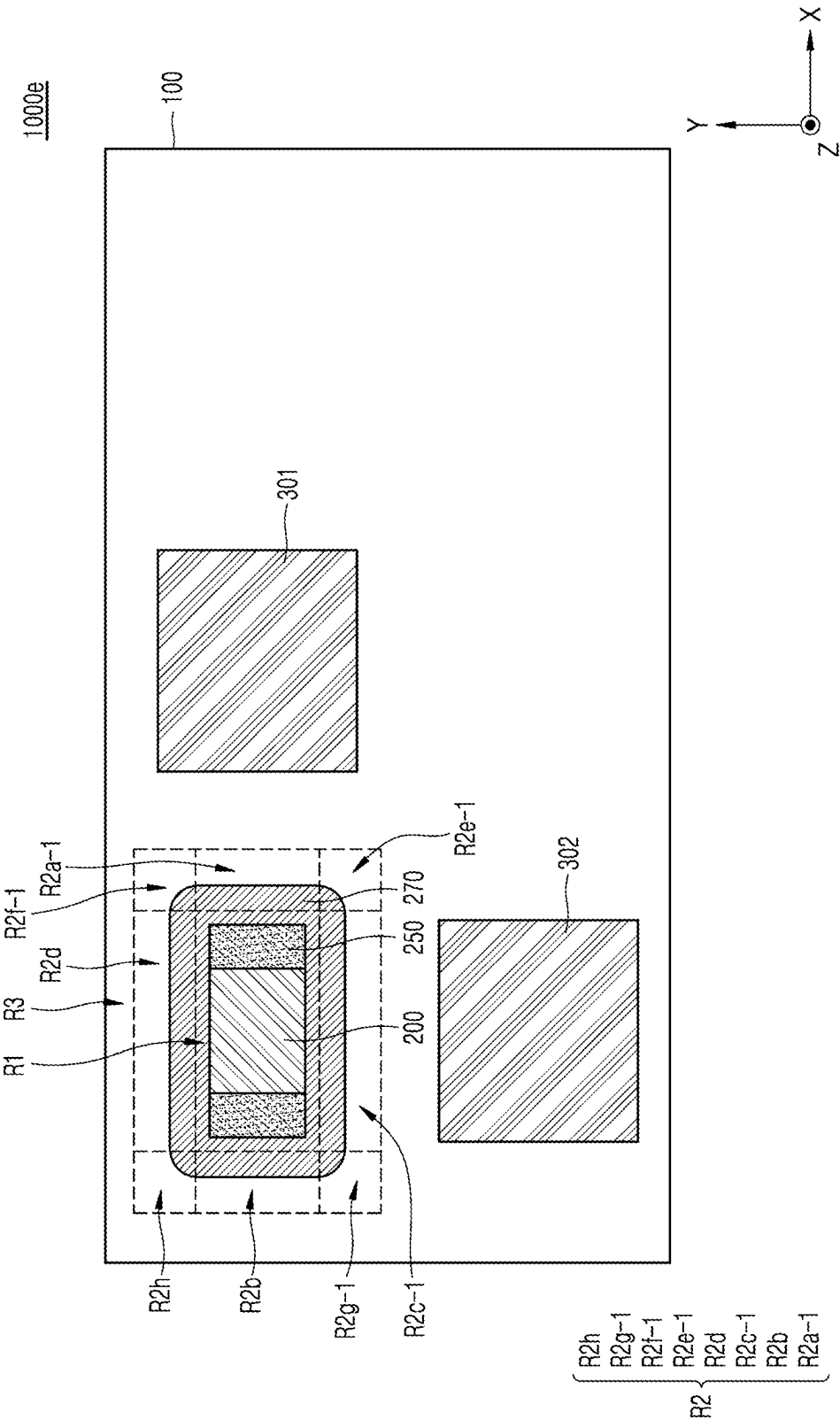


FIG. 7

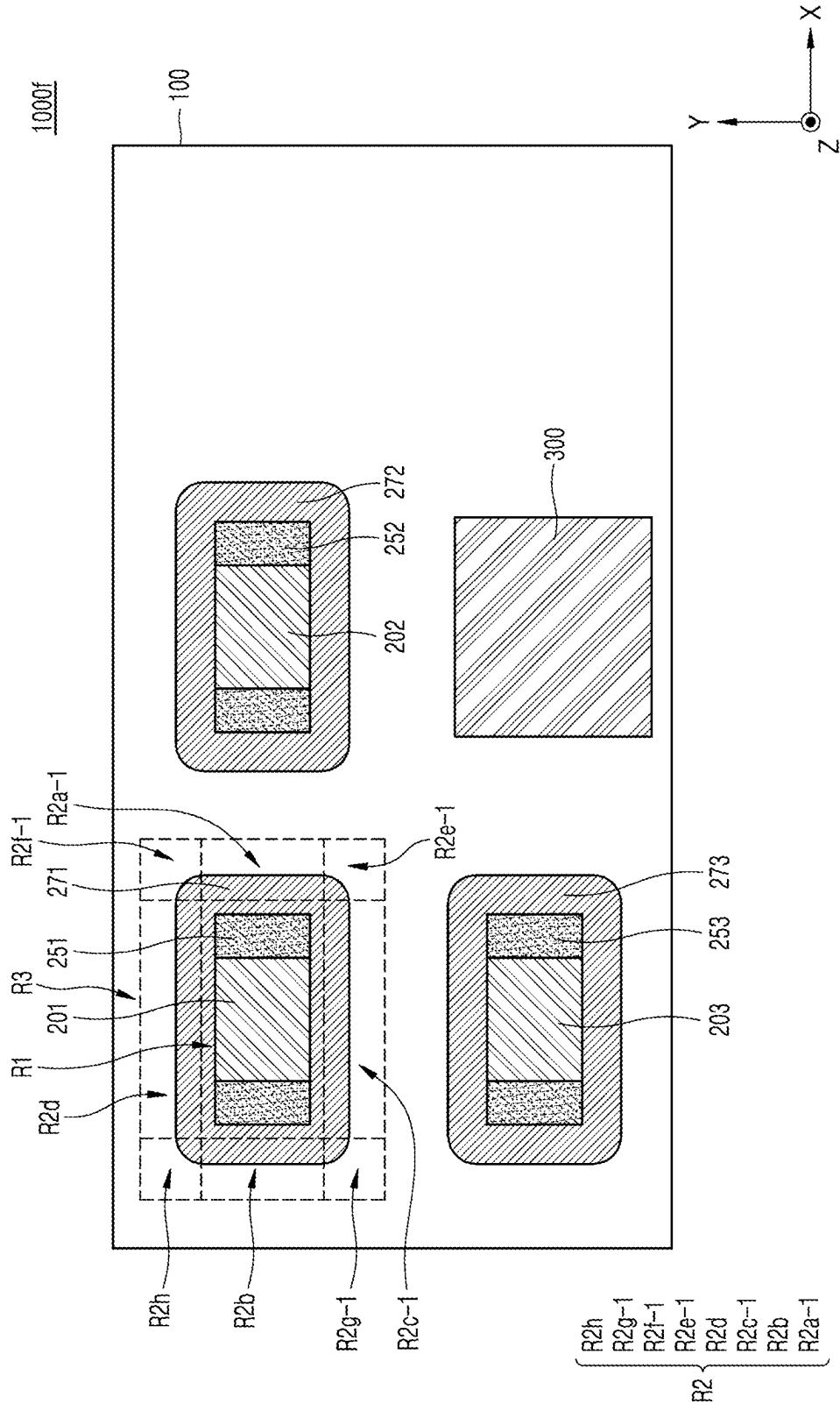


FIG. 8

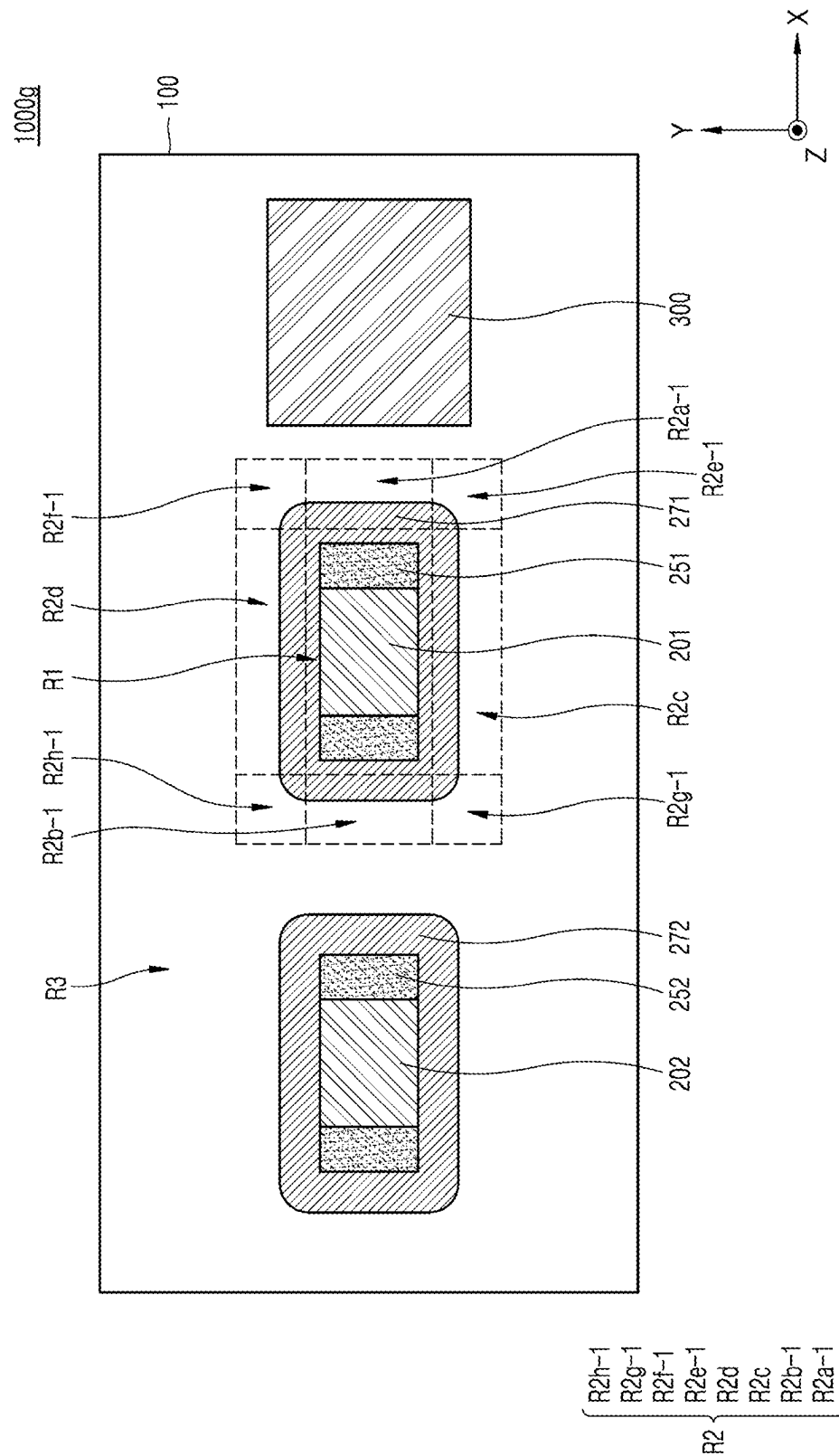


FIG. 10

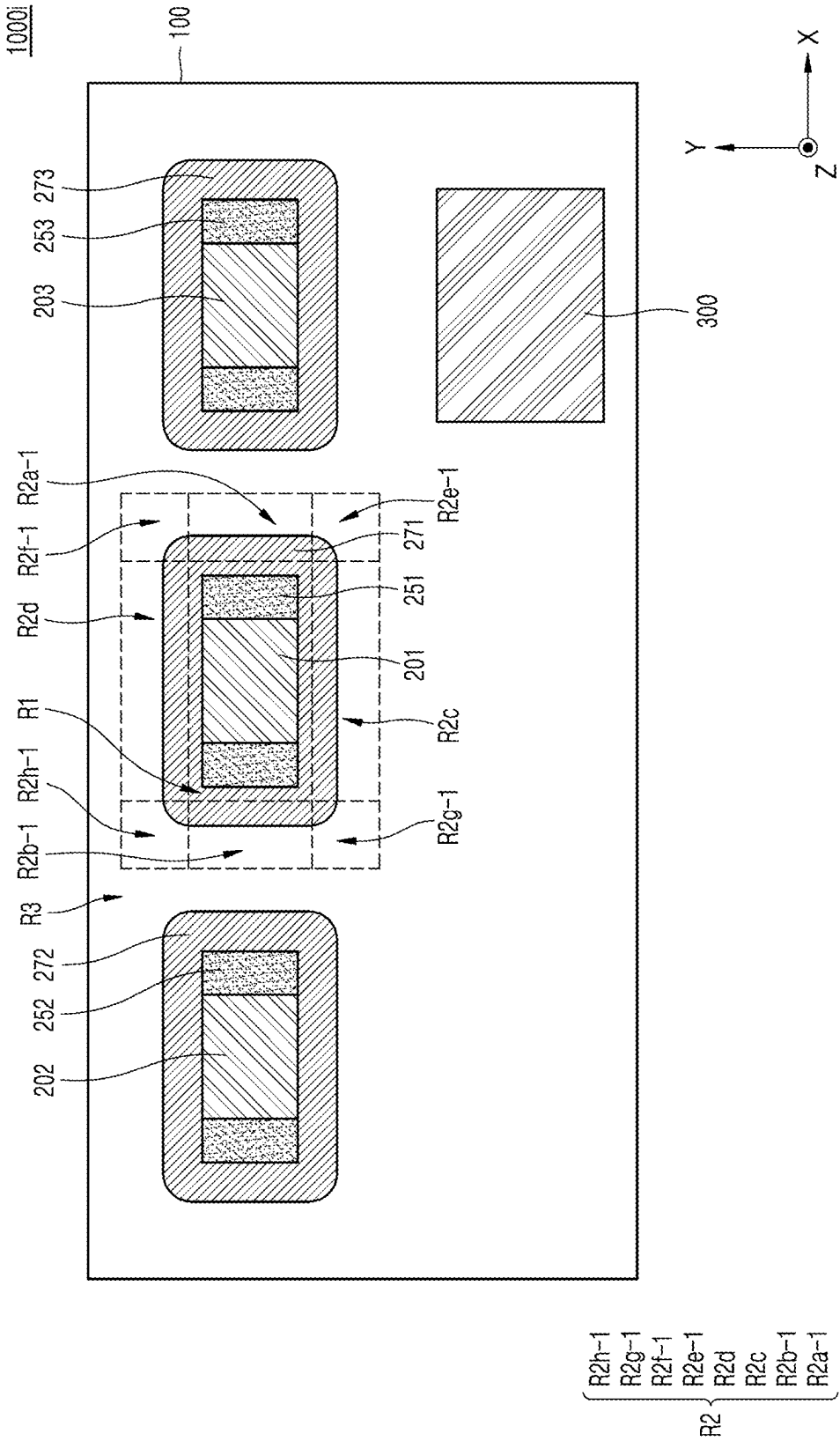


FIG. 11

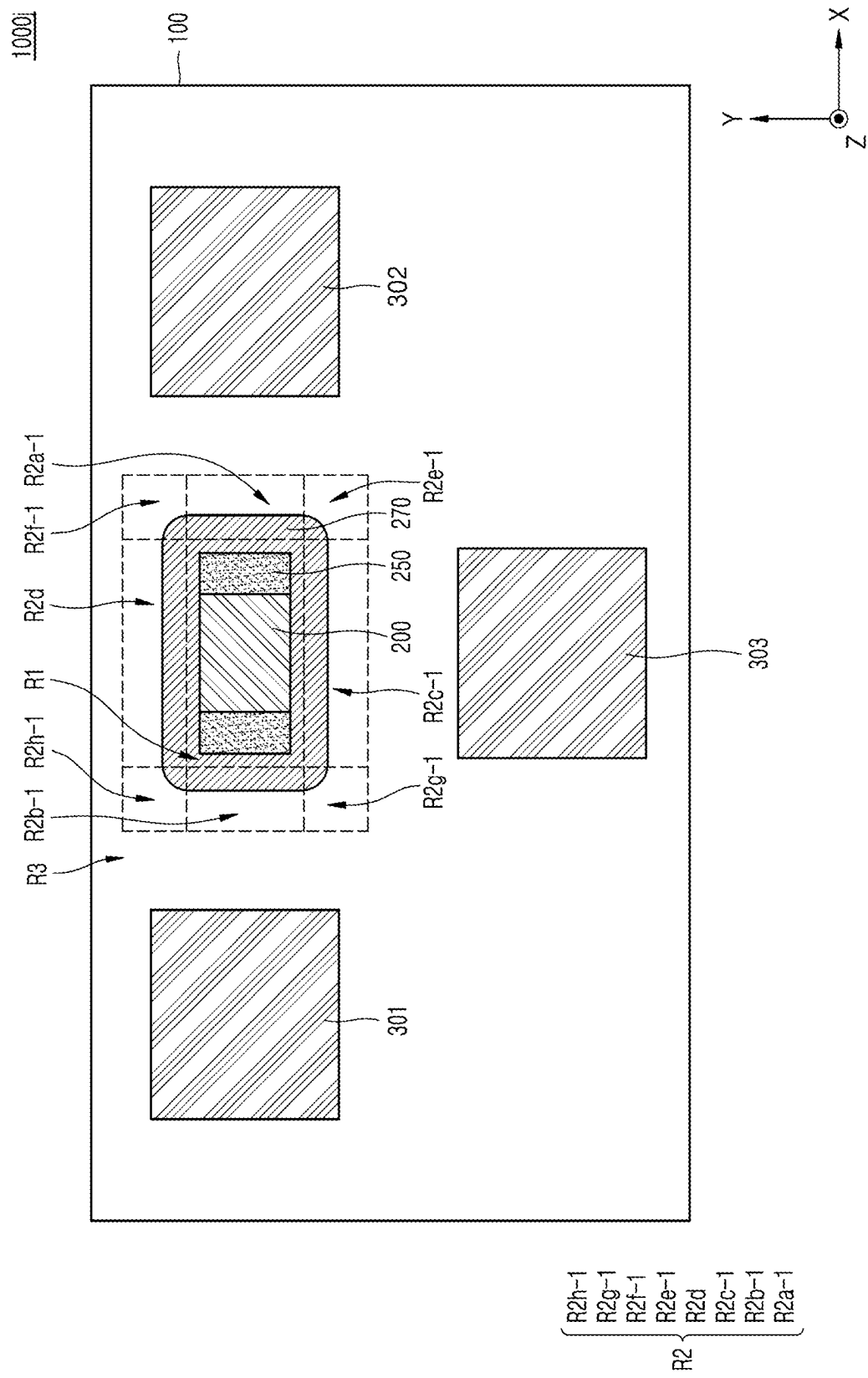


FIG. 12

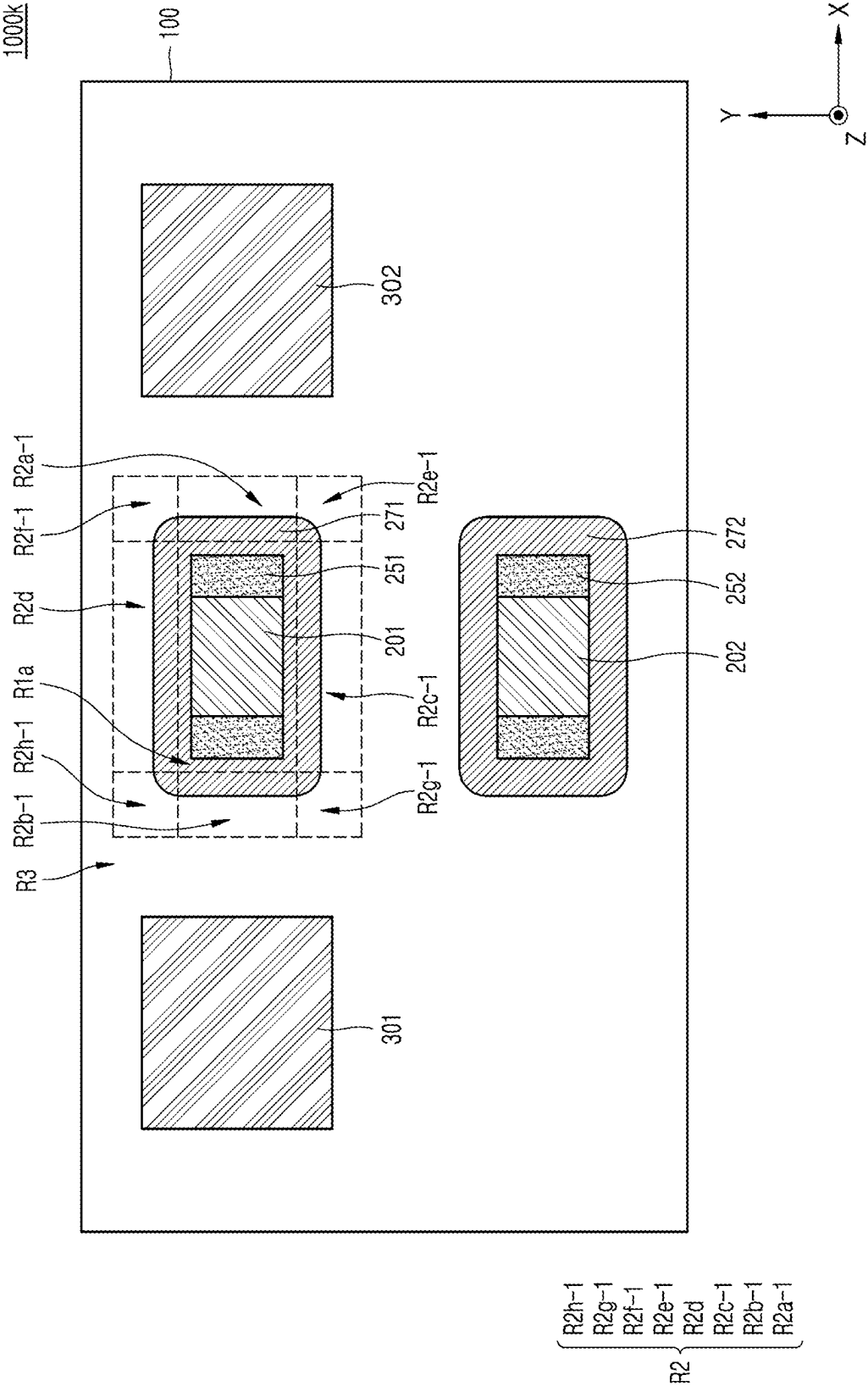


FIG. 13

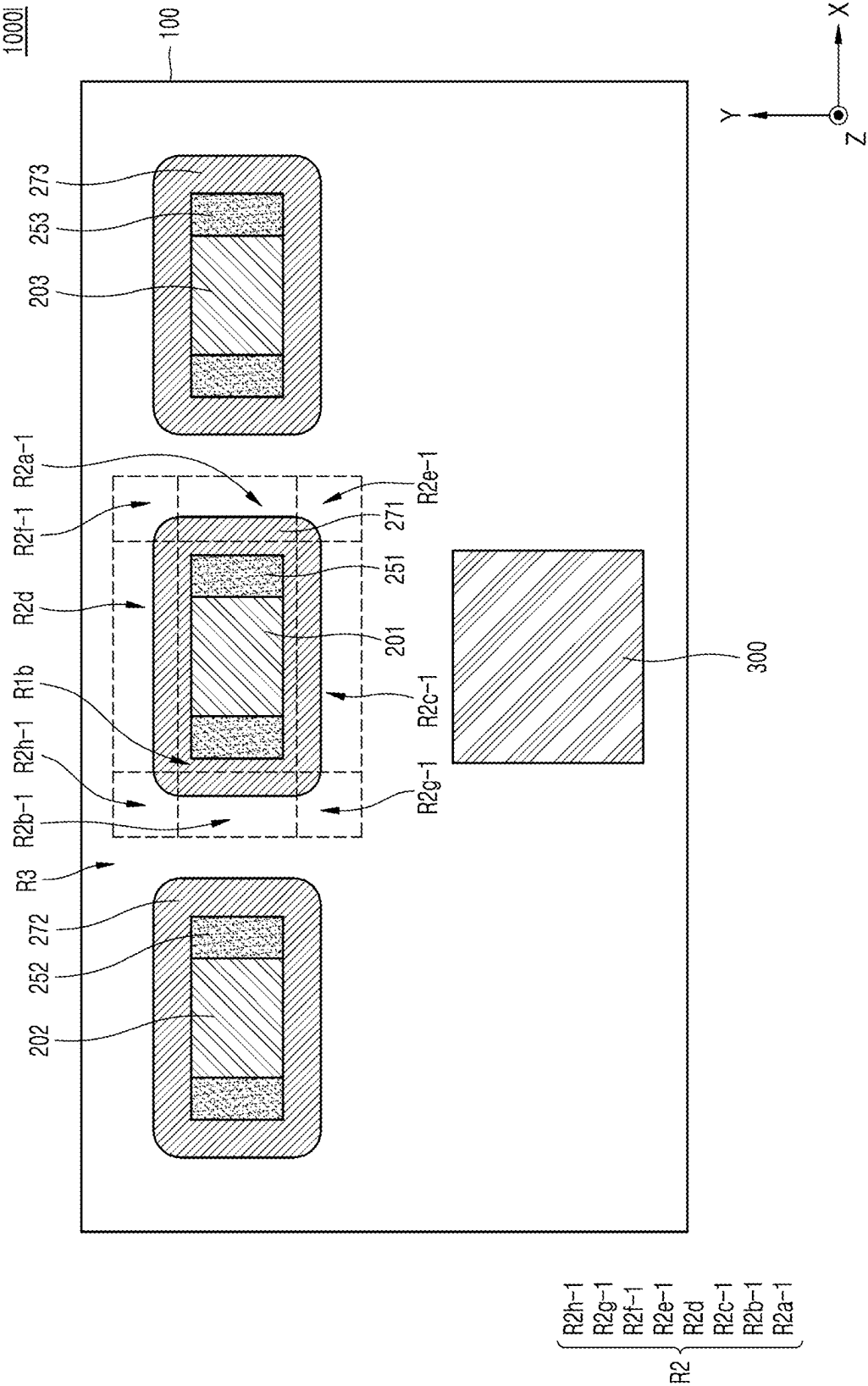


FIG. 14

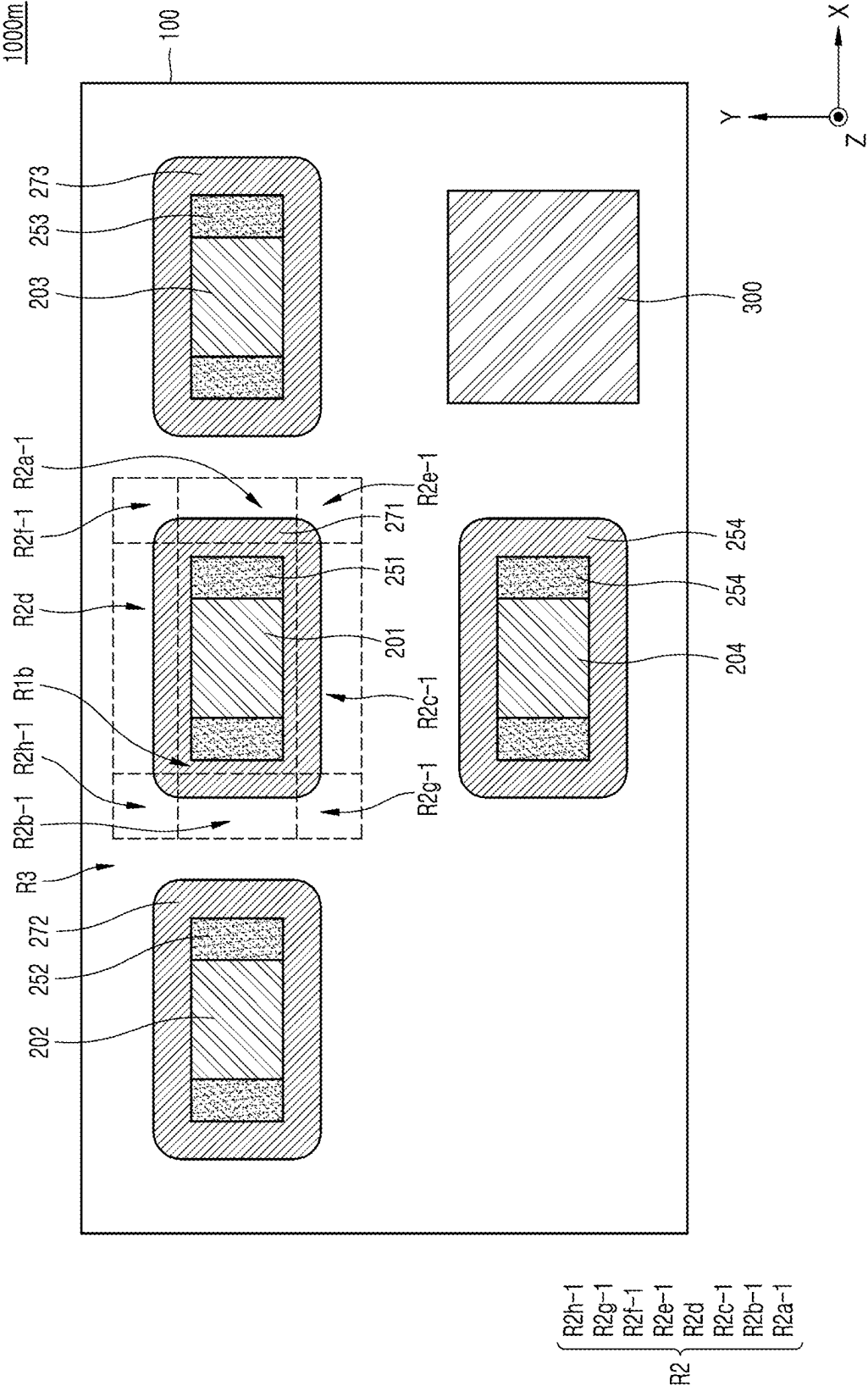


FIG. 15

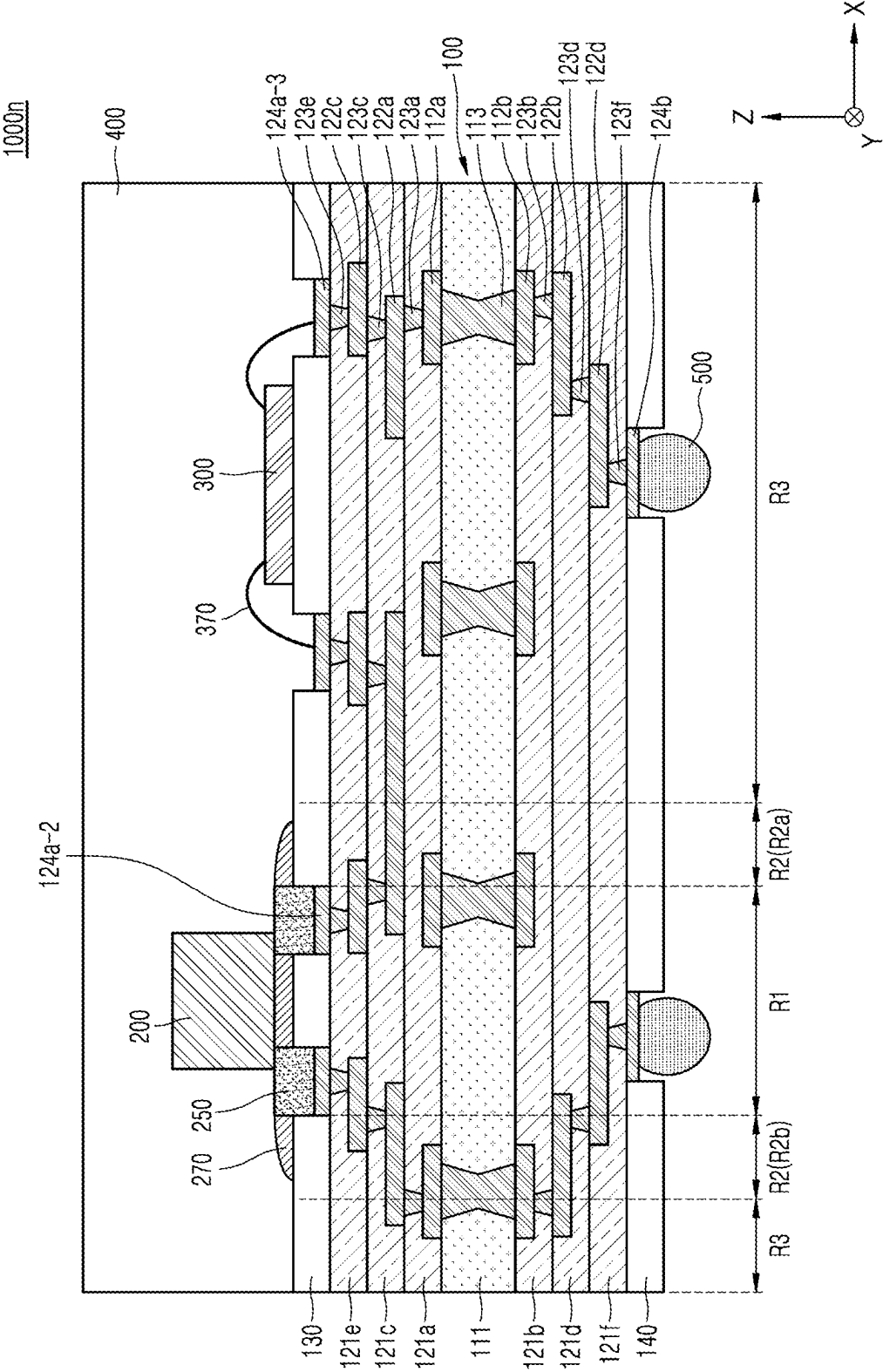


FIG. 16

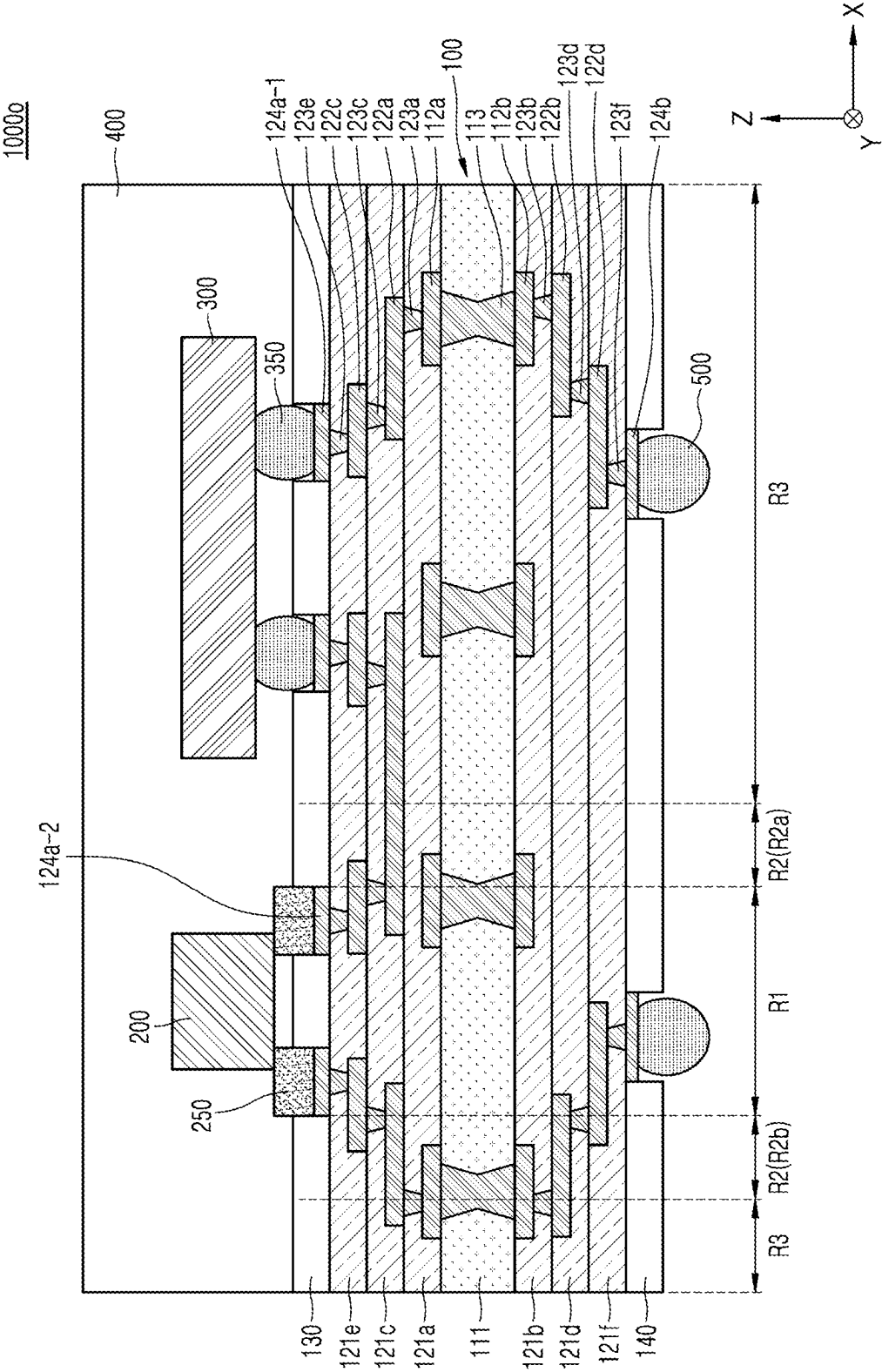


FIG. 17

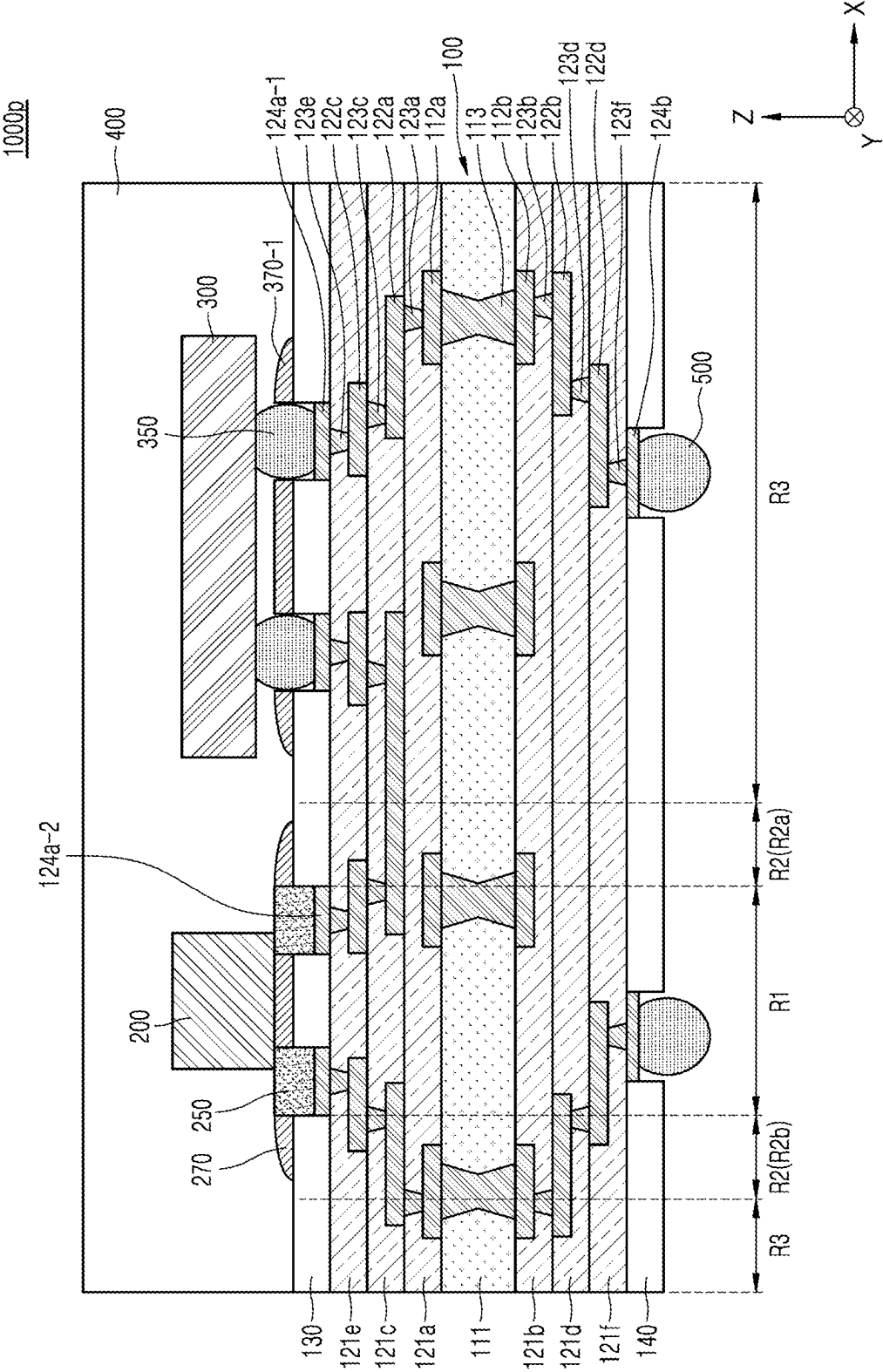


FIG. 18

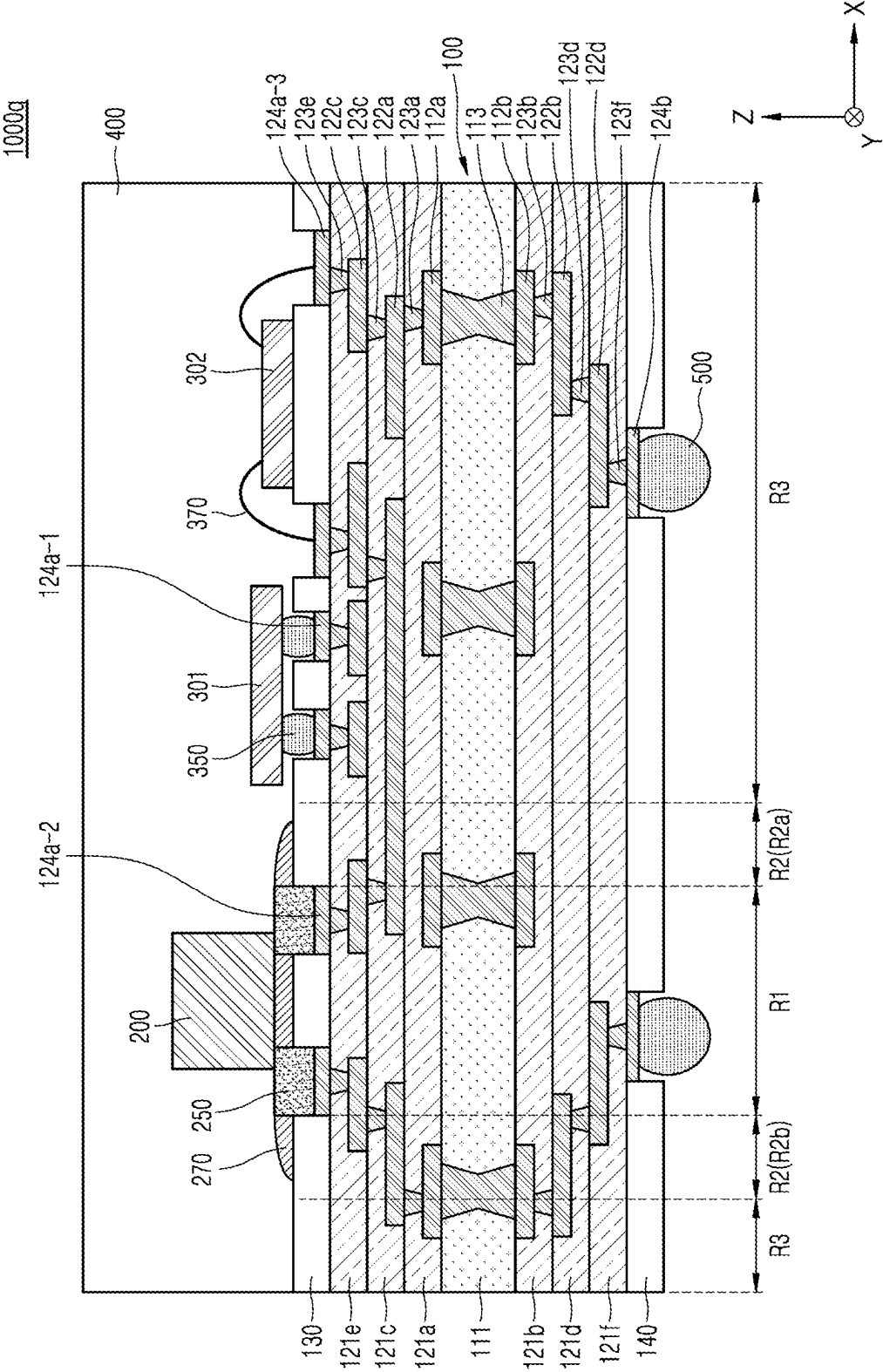


FIG. 20A

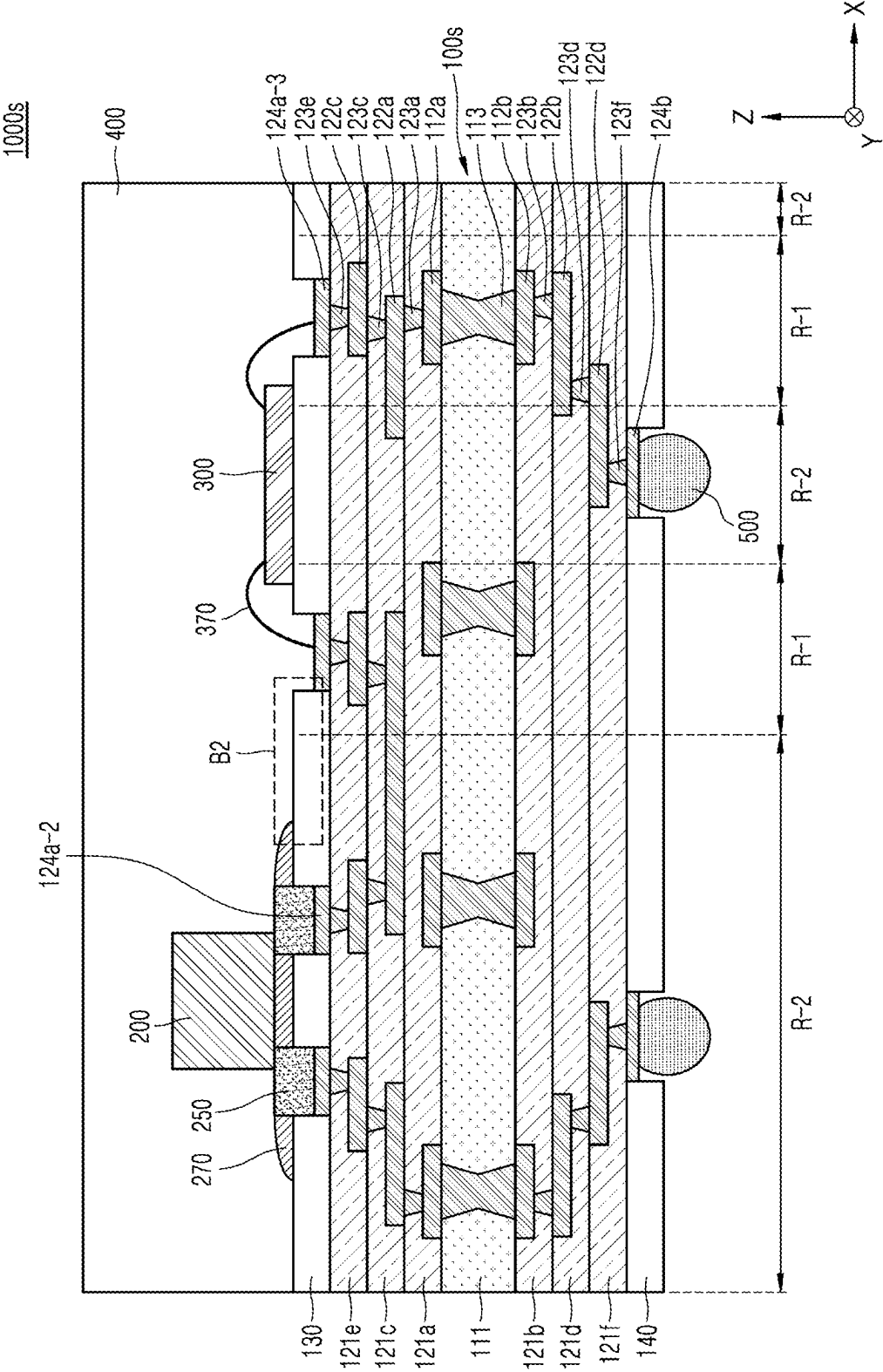


FIG. 20B

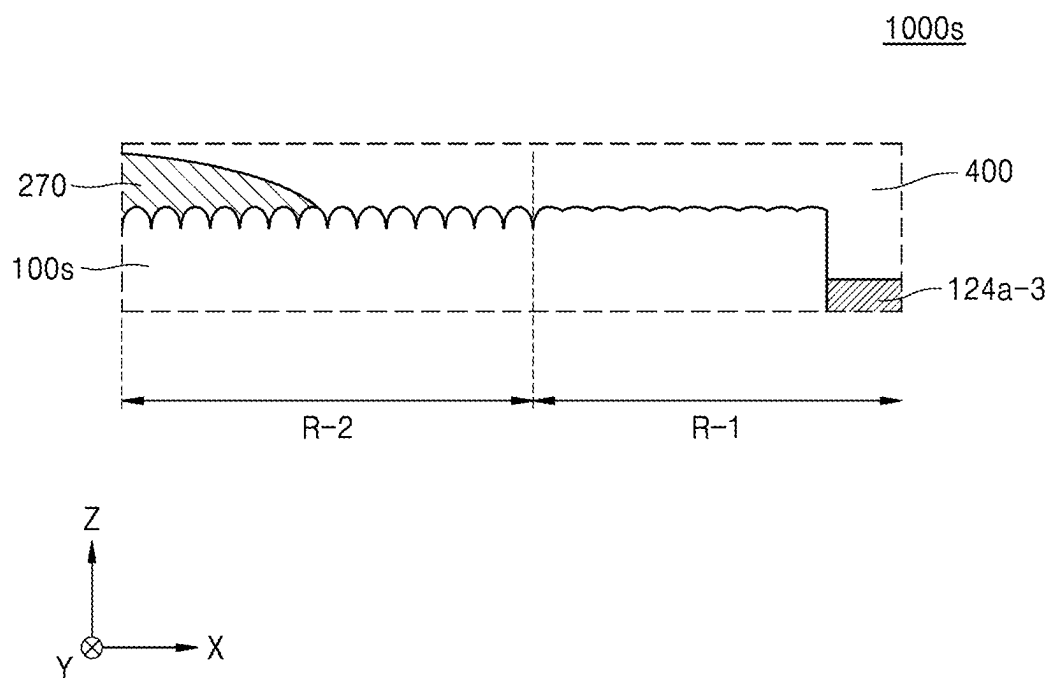
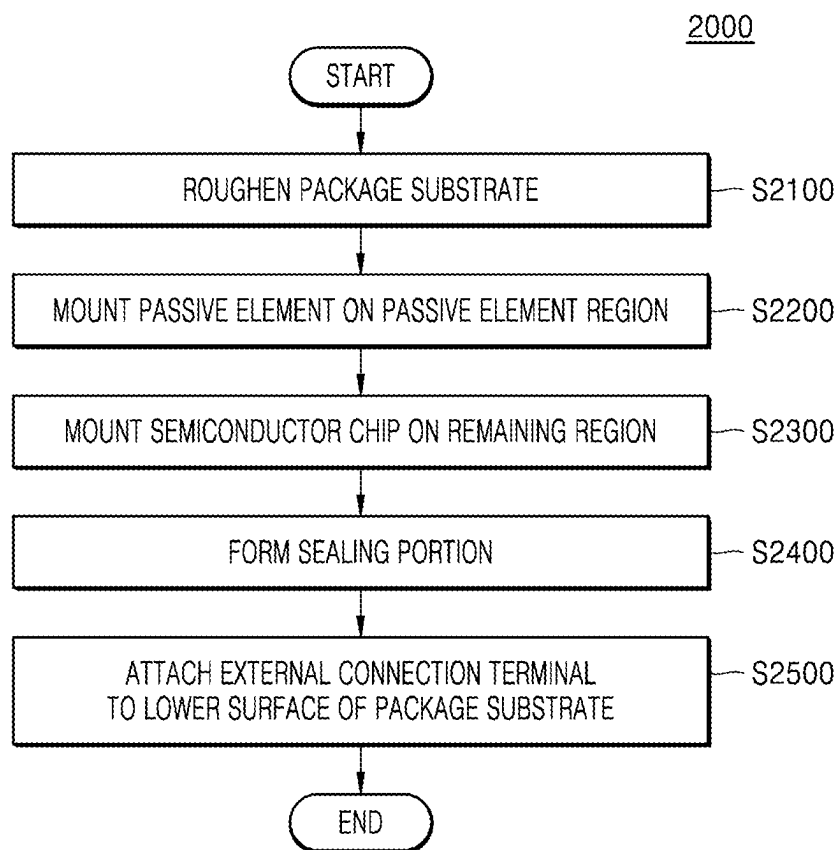


FIG. 21



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SEMICONDUCTOR PACKAGE**CROSS-REFERENCE TO RELATED APPLICATION**

This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0134436, filed on Oct. 8, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND**1. Field**

Embodiments relate to a semiconductor package. In particular, embodiments relate to a semiconductor package including a passive element and a semiconductor chip.

2. Description of the Related Art

A solder paste may be used when a passive element is mounted on a package substrate. A passive element connecting member between the package substrate and the passive element and a flux layer on the package substrate may be made from the solder paste.

SUMMARY

According to an aspect of embodiments, there is provided a semiconductor package that may include a package substrate having a passive element region, a passive element adjacent region adjacent to the passive element region, and a remaining region; a first passive element on an upper surface of the passive element region; a first semiconductor chip on an upper surface of the remaining region; and a sealing portion covering the package substrate, the first passive element, and the first semiconductor chip, wherein the passive element adjacent region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side of the first passive element, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side of the first passive element, and a roughness of an upper surface of at least one of the first sub-region to the fourth sub-region is greater than the roughness of an upper surface of the remaining region.

According to another aspect of embodiments, there is provided a semiconductor package that may include a package substrate having a wire pad, a wire pad, a wire pad adjacent region adjacent to the wire pad, and a remaining region; a passive element on an upper surface of the remaining region; a semiconductor chip on the upper surface of the remaining region; a wire electrically connecting the semiconductor chip to the wire pad; and a sealing portion covering the package substrate, the passive element, the semiconductor chip, and the wire, wherein the roughness of an upper surface of the wire pad adjacent region is less than the roughness of the upper surface of the remaining region.

According to another aspect of embodiments, there is provided a semiconductor package that may include a package substrate having a passive element region, a passive element adjacent region adjacent to the passive element region, and a remaining region; a plurality of external connection terminals on a lower surface of the package substrate; a passive element on an upper surface of the passive element region; a memory controller chip on the

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upper surface of the remaining region; a chip bump connecting the memory controller chip to the package substrate; a memory chip on the memory controller chip or the remaining region; a wire connecting the memory chip to the package substrate; and a sealing portion covering the package substrate, the passive element, the memory controller chip, the chip bump, the memory chip, and the wire, wherein the passive element adjacent region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side of the first passive element, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side of the first passive element, and the roughness of the upper surface of at least one of the first sub-region to the fourth sub-region is greater than the roughness of the upper surface of the remaining region.

BRIEF DESCRIPTION OF THE DRAWINGS

Features will become apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings, in which:

FIG. 1A is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 1B is an enlarged view of region B in FIG. 1A;

FIG. 1C is a plan view of a semiconductor package according to an embodiment;

FIG. 2 is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 3 is a plan view of a semiconductor package according to an embodiment;

FIG. 4 is a plan view of a semiconductor package according to an embodiment;

FIG. 5 is a plan view of a semiconductor package according to an embodiment;

FIG. 6 is a plan view of a semiconductor package according to an embodiment;

FIG. 7 is a plan view of a semiconductor package according to an embodiment;

FIG. 8 is a plan view of a semiconductor package according to an embodiment;

FIG. 9 is a plan view of a semiconductor package according to one embodiment;

FIG. 10 is a plan view of a semiconductor package according to an embodiment;

FIG. 11 is a plan view of a semiconductor package according to an embodiment;

FIG. 12 is a plan view of a semiconductor package according to an embodiment;

FIG. 13 is a plan view of a semiconductor package according to an embodiment;

FIG. 14 is a plan view of a semiconductor package according to an embodiment;

FIG. 15 is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 16 is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 17 is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 18 is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 19 is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 20A is a cross-sectional view of a semiconductor package according to an embodiment;

FIG. 20B is an enlarged view of region B2 in FIG. 20A; and

FIG. 21 is a flowchart of a method of manufacturing a semiconductor package, according to an embodiment.

DETAILED DESCRIPTION

FIG. 1A is a cross-sectional view showing a semiconductor package 1000 according to an embodiment. FIG. 1B is an enlarged view of region B in FIG. 1A, and FIG. 1C is a plan view showing the semiconductor package 1000 according to an embodiment.

Referring to FIGS. 1A to 1C, the semiconductor package 1000 may include a package substrate 100, a plurality of external connection terminals 500 on a lower surface of the package substrate 100, a passive element 200 on an upper surface of the package substrate 100, a plurality of passive element connecting members 250 for connecting the passive element 200 to the package substrate 100, a semiconductor chip 300 on the upper surface of the package substrate 100, a plurality of chip bumps 350 connecting the semiconductor chip 300 to the package substrate 100, and a sealing portion 400 covering the semiconductor chip 300. In some embodiments, the semiconductor package 1000 may further include a flux layer 270 on the package substrate 100.

The package substrate 100 may include a printed circuit board (PCB). For example, the package substrate 100 may include a core layer 111, a first insulating layer 121a on an upper surface of the core layer 111, a second insulating layer 121b on a lower surface of the core layer 111, a third insulating layer 121c on the upper surface of the first insulating layer 121a, a fourth insulating layer 121d on the lower surface of the second insulating layer 121b, a fifth insulating layer 121e on the upper surface of the third insulating layer 121c, and a sixth insulating layer 121f on the lower surface of the fourth insulating layer 121d. The package substrate 100 may further include a first protective layer 130 on an upper surface of the fifth insulating layer 121e and a second protective layer 140 on a lower surface of the sixth insulating layer 121f.

The package substrate 100 may further include a core via 113 penetrating the core layer 111. The package substrate 100 may further include a first core wiring layer 112a positioned on the upper surface of the core layer 111 and in contact with the core via 113. The package substrate 100 may further include a second core wiring layer 112b positioned on the lower surface of the core layer 111 and in contact with the core via 113.

The package substrate 100 may further include a first wiring layer 122a on the upper surface of the first insulating layer 121a. The package substrate 100 may further include a first via layer 123a passing through the first insulating layer 121a between the first wiring layer 122a and the first core wiring layer 112a. The package substrate 100 may further include a second wiring layer 122b on a lower surface of the second insulating layer 121b. The package substrate 100 may further include a second via layer 123b passing through the second insulating layer 121b between the second wiring layer 122b and the second core wiring layer 112b. The package substrate 100 may further include a third wiring layer 122c on the upper surface of the third insulating layer 121c. The package substrate 100 may further include a third via layer 123c passing through the third insulating layer 121c between the third wiring layer 122c and the first wiring layer 122a. The package substrate 100 may further include a fourth wiring layer 122d on the lower surface of the fourth insulating layer 121d. The package substrate 100 may further include a fourth via layer 123d

passing through the fourth insulating layer 121d between the fourth wiring layer 122d and the second wiring layer 122b.

The package substrate 100 may further include a plurality of bump pads 124a-1 and a plurality of passive element pads 124a-2 on the upper surface of the fifth insulating layer 121e. The first protective layer 130 may have a plurality of openings exposing the plurality of bump pads 124a-1 and the plurality of passive element pads 124a-2, respectively. The package substrate 100 may further include a fifth via layer 123e passing through the fifth insulating layer 121e between the third wiring layer 122c and the plurality of bump pads 124a-1 and between the third wiring layer 122c and the plurality of passive element pads 124a-2. The package substrate 100 may further include a plurality of external connection terminal pads 124b on the lower surface of the sixth insulating layer 121f. The second protective layer 140 may have a plurality of openings exposing the plurality of external connection terminal pads 124b. The package substrate 100 may further include a sixth via layer 123f passing through the sixth insulating layer 121f between the fourth wiring layer 122d and the plurality of external connection terminal pads 124b.

In FIG. 1A, the package substrate 100 includes six insulating layers (i.e., the first to sixth insulating layers 121a to 121f), four wiring layers (i.e., the first to fourth wiring layers 122a to 122d), and six via layers (i.e., the first to sixth via layers 123a to 123f). However, the numbers of insulating layers, wiring layers, and via layers included in the package substrate 100 may be variously modified.

For example, the core layer 111 may include an insulating material, a thermosetting resin (e.g., an epoxy resin) or a thermoplastic resin (e.g., polyimide). The core layer 111 may include a material including a reinforcing material, e.g., fiberglass, and/or an inorganic filler, e.g., a copper clad laminate (CCL) or an unclad CCL. The core layer 111 may include a metal plate, a glass plate, or a ceramic plate.

The first to sixth insulating layers 121a to 121f may include a thermosetting resin, e.g., epoxy, or a thermoplastic resin, e.g., polyimide. In some embodiments, the first to sixth insulating layers 121a to 121f may include a material including a reinforcing material, e.g., glass fiber, and/or an inorganic filler in addition to a thermoplastic resin and/or a thermosetting resin, e.g., prepreg or Ajinomoto build-up film (ABF). In some embodiments, the first protective layer 130 and the second protective layer 140 may include solder resist.

The core via 113, the first and second core wiring layers 112a and 112b, the first to fourth wiring layers 122a to 122d, the first to sixth via layers 123a to 123f, the bump pads 124a-1, the passive element pads 124a-2, and the external connection terminal pads 124b may include a metal material. The metal material may include, e.g., copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or an alloy thereof.

As shown in FIG. 1C, the package substrate 100 may have a passive element region R1, a passive element adjacent region R2 adjacent to the passive element region R1, and a remaining region R3. For example, the passive element adjacent region R2 may be a peripheral region of the passive element region R1. In some embodiments, the passive element adjacent region R2 may surround, e.g., an entire perimeter of, the passive element region R1, and the remaining region R3 may surround, e.g., an entire perimeter of, the passive element adjacent region R2, as viewed in a top view (FIG. 1C).

The plurality of external connection terminals 500 may be respectively positioned on the plurality of external connec-

tion terminal pads **124b** of the package substrate **100**. Each of the external connection terminals **500** may include, e.g., tin (Sn) or a tin (Sn) alloy. In some embodiments, each external connection terminal **500** may be formed of a solder ball.

The passive element **200** may be located on the passive element region **R1**. The passive element **200** may be positioned on the plurality of passive element pads **124a-2** of the package substrate **100**. The passive element **200** may be, e.g., a capacitor, an inductor, or a resistor.

The plurality of passive element connecting members **250** may be located between the passive element region **R1** of the package substrate **100** and the passive element **200**, e.g., the plurality of passive element connecting members **250** may be between the package substrate **100** and the passive element **200** along the Z direction. The plurality of passive element connecting members **250** may be located between the passive element **200** and the plurality of passive element pads **124a-2**, respectively. The plurality of passive element connecting members **250** may include, e.g., tin (Sn) or tin (Sn) alloys. In some embodiments, a plurality of passive element connecting members **250** may be formed of a solder paste.

The semiconductor chip **300** may be located on the remaining region **R3**. The semiconductor chip **300** may be located on the plurality of bump pads **124a-1** of the package substrate **100**. The semiconductor chip **300** may include an arbitrary kind of integrated circuit including, e.g., a memory circuit, a logic circuit, or a combination thereof. The memory circuit may be, e.g., a dynamic random access memory (DRAM) circuit, a static random access memory (SRAM) circuit, a flash memory circuit, an electrically erasable and programmable read-only memory (EEPROM) circuit, a phase-change random access memory (PRAM) circuit, a magnetic random access memory (MRAM) circuit, a resistive random access memory (RRAM) circuit, or a combination thereof. The logic circuit may be, e.g., a central processing unit (CPU) circuit, a graphics processing unit (GPU) circuit, a memory controller circuit, an application specific integrated circuit (ASIC) circuit, an application processor (AP) circuit, or a combination thereof.

The chip bumps **350** may be located between the remaining region **R3** of the package substrate **100** and the semiconductor chip **300**, e.g., the chip bumps **350** may be between the package substrate **100** and the semiconductor chip **300** along the Z direction. The chip bumps **350** may be located between the semiconductor chip **300** and the plurality of bump pads **124a-1**, respectively. The chip bumps **350** may include, e.g., a tin (Sn) or a tin (Sn) alloy. In some embodiments, the chip bump **350** may be formed of a solder ball.

The sealing portion **400** may cover the package substrate **100**, the passive element **200**, and the semiconductor chip **300**. The sealing portion **400** may include an epoxy resin. For example, the sealing portion **400** may include an epoxy mold compound (EMC).

The flux layer **270** may be located on the passive element adjacent region **R2**. In some embodiments, the flux layer **270** may be further positioned on the passive element region **R1**. In some embodiments, the flux layer **270** may be formed of flux of solder paste.

Referring to FIG. 1C, the passive element adjacent region **R2** may include a first sub-region **R2a** on the first side of the passive element **200**, a second sub-region **R2b** on a second side opposite to the first side of the passive element **200**, a third sub-region **R2c** on a third side of the passive element **200**, and a fourth sub-region **R2d** of a fourth side opposite

to the third side of the passive element **200**. In some embodiments, the passive element adjacent region **R2** may further include a fifth sub-region **R2e** in which the first sub-region **R2a** intersects with the third sub-region **R2c**, a sixth sub-region **R2f** in which the first sub-region **R2a** intersects with the fourth sub-region **R2d**, a seventh sub-region **R2g** in which the third sub-region **R2c** intersects with the second sub-region **R2b**, and an eighth sub-region **R2h** in which the fourth sub-region **R2d** intersects with the second sub-region **R2b**. In some embodiments, e.g., each of, the width W_a of the first sub-region **R2a**, the width W_b of the second sub-region **R2b**, the width W_c of the third sub-region **R2c**, and the width W_d of the fourth sub-region **R2d** may be about 10 μm to about 500 μm .

In some embodiments, the roughness of the upper surface of the first sub-region **R2a** to the fourth sub-region **R2d** may be greater than the roughness of the upper surface of the remaining region **R3**. For example, referring to FIG. 1B, the roughness of the upper surface, i.e., a surface facing the passive element **200**, of the first sub-region **R2a** to the fourth sub-region **R2d** may be greater than the roughness of the upper surface, i.e., a surface facing the semiconductor chip **300**, of the remaining region **R3**.

In this specification, roughness means average roughness (roughness average, R_a). In some embodiments, the roughness of the upper surface of the first sub-region **R2a** to the fourth sub-region **R2d** may be greater than the roughness of the upper surface of the remaining region **R3** by about 80 nm or more. In some embodiments, the roughness of the upper surface of the fifth sub-region **R2e** to the eighth sub-region **R2h** may be greater than the roughness of the upper surface of the remaining region **R3**. In some embodiments, the roughness of the upper surface of the fifth sub-region **R2e** to the eighth sub-region **R2h** may be greater than the roughness of the upper surface of the remaining region **R3** by about 80 nm or more.

The greater the roughness of the surface through which the flux flows, the wider the flux may flow. Because the passive element adjacent region **R2** has a rough surface, the flux layer **270** may be formed wider on the passive element adjacent region **R2**. Accordingly, the concentration of the flux mixed in the sealing portion **400** may be reduced. Therefore, the possibility of delamination caused by combining the sealing portion **400** with the flux layer **270** may be reduced. On the other hand, by forming the remaining region **R3** to be relatively flat, i.e., a lower surface roughness, the flow of the flux layer **270** to the remaining region **R3** may be reduced. Therefore, the flux layer **270** may be prevented from contaminating the bump pads **124a-1** located in the remaining region **R3**.

In some embodiments, the roughness of the upper surface of the passive element region **R1** may be greater than the roughness of the upper surface of the remaining region **R3**. For example, the roughness of the upper surface of the passive element region **R1** may be greater than the roughness of the upper surface of the remaining region **R3** by about 80 nm or more.

FIG. 2 is a cross-sectional view showing a semiconductor package **1000a** according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIG. 1A to 1C and the semiconductor package **1000a** shown in FIG. 2 are described.

Referring to FIG. 2, the roughness of the upper surface of the passive element region **R1** may be less than the roughness of the upper surface of the passive element adjacent region **R2**. For example, the roughness of the upper surface of the passive element region **R1** may be less than the

roughness of the upper surface of the first sub-region **R2a**. For example, the roughness of the upper surface of the passive element region **R1** may be less than the roughness of the upper surface of the passive element adjacent region **R2** by about 80 nm or more. The roughness of the upper surface of the passive element region **R1** may be less than the roughness of the upper surface of the second to fourth sub-regions (refer to **R2b** to **R2d**, and FIG. 1C). For example, the roughness of the upper surface of the passive element region **R1** may be less than the roughness of the upper surface of the upper surface of the second to fourth sub-regions **R2b** to **R2d** (refer to FIG. 1C) by about 80 nm or more.

FIG. 3 is a plan view showing a semiconductor package **1000b** according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIGS. 1A to 1C and the semiconductor package **1000b** shown in FIG. 3 are described.

Referring to FIG. 3, the roughness of an upper surface of a first sub-region **R2a-1** may be less than the roughness of an upper surface of a second sub-region **R2b** to a fourth sub-region **R2d**. For example, the roughness of the upper surface of the first sub-region **R2a-1** may be less than the roughness of the upper surface of the second sub-region **R2b** to the fourth sub-region **R2d** by 80 nm or more.

In some embodiments, the roughness of upper surfaces of a fifth sub-region **R2e-1** and a sixth sub-region **R2f-1** may be less than roughness of the upper surfaces of a seventh sub-region **R2g** and an eighth sub-region **R2h**. For example, the roughness of the upper surface of the fifth sub-region **R2e-1** and the sixth sub-region **R2f-1** may be less than the roughness of the upper surface of the seventh sub-region **R2g** and the eighth sub-region **R2h** by 80 nm or more.

By forming the first sub-region **R2a-1**, the fifth sub-region **R2e-1**, and the sixth sub-region **R2f-1** to be relatively flat, the flow of the flux layer **270** to the first side of the passive element **200** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the bump pad **124a-1** (refer to FIG. 1A) under the semiconductor chip **300** positioned on the first side of the passive element **200**.

FIG. 4 is a plan view showing a semiconductor package **1000c** according to an embodiment. Hereinafter, differences between the semiconductor packages **1000** shown in FIGS. 1A to 1C and the semiconductor package **1000c** shown in FIG. 4 are described.

Referring to FIG. 4, the semiconductor package **1000c** may include the package substrate **100**, a first passive element **201**, a first passive element connecting member **251**, a first flux layer **271**, a second passive element **202**, a second passive element connecting member **252**, a second flux layer **272**, and the semiconductor chip **300**. The package substrate **100** may include a first passive element region **R1a**, a first passive element adjacent region **R2** adjacent to the first passive element region **R1a**, a second passive element region **R1b** positioned on the first side of the first passive element region **R1a**, a second passive element adjacent region **R4** adjacent to the second passive element region **R1b**, and a remaining region **R3**. The first passive element **201** may be located on the first passive element region **R1a**, and the second passive element **202** may be located on the second passive element region **R1b**, and the semiconductor chip **300** may be located on the remaining region **R3**. A plurality of first passive element connecting members **251** may connect the first passive element **201** to the package substrate **100**, and a plurality of second passive element connecting members **252** may connect the second passive element **202** to the package substrate **100**. The first flux layer **271** may be located on the first passive element

region **R1a** and the first passive element adjacent region **R2**, and the second flux layer **272** may be located on the second passive element region **R1b** and the second passive element adjacent region **R4**.

The first passive element adjacent region **R2** may include a first sub-region **R2a-1** on a first side of the first passive element **201**, a second sub-region **R2b** on a second side opposite to the first side of the first passive element **201**, a third sub-region **R2c** on a third side of the first passive element **201**, and the fourth sub-region **R2d** on a fourth side opposite to the third side of the first passive element **201**. In some embodiments, the first passive element adjacent region **R2** may further include a fifth sub-region **R2e-1** in which the first sub-region **R2a-1** intersects with the third sub-region **R2c**, a sixth sub-region **R2f-1** in which the first sub-region **R2a-1** intersects with the fourth sub-region **R2d**, a seventh sub-region **R2g** in which the third sub-region **R2c** intersects with the second sub-region **R2b**, and an eighth sub-region **R2h** in which the fourth sub-region **R2d** intersects with the second sub-region **R2b**.

The second passive element **202** may be located on the first side of the first passive element **201**, e.g., the second passive element **202** may be between the first passive element **201** and the semiconductor chip **300** along the X direction. In this case, the roughness of the upper surface of the first sub-region **R2a-1** of the first passive element adjacent region **R2** may be less than the roughness of the upper surfaces of the second sub-region **R2b** to the fourth sub-region **R2d** of the first passive element adjacent region **R2**. For example, the roughness of the upper surface of the first sub-region **R2a-1** of the first passive element adjacent region **R2** may be less than that of the upper surface of the second sub-region **R2b** to the fourth sub-region **R2d** of the first passive element adjacent region **R2** by 80 nm or more.

The roughness of the upper surfaces of the fifth sub-region **R2e-1** and the sixth sub-region **R2f-1** of the first passive element adjacent region **R2** may be less than the roughness of the upper surfaces of the seventh sub-region **R2g** and the eighth sub-region **R2h** of the first passive element adjacent region **R2**. For example, the roughness of upper surfaces of the fifth sub-region **R2e-1** and the sixth sub-region **R2f-1** of the first passive element adjacent region **R2** may be less than the roughness of upper surfaces of the seventh sub-region **R2g** and the eighth sub-region **R2h** of the first passive element adjacent region **R2** by 80 nm or more.

By forming the first sub-region **R2a-1**, the fifth sub-region **R2e-1**, and the sixth sub-region **R2f-1** of the first passive element adjacent region **R2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the second flux layer **272**. Therefore, it is possible to prevent an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. 1A). Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. 1A) and the package substrate **100**.

The second passive element adjacent region **R4** may include a first sub-region **R4a-1** on a first side of the second passive element **202**, a second sub-region **R4b-1** on a second side opposite to the first side of the second passive element **202**, a third sub-region **R4c** on a third side of the second passive element **202**, and a fourth sub-region **R4d** on a fourth side opposite to the third side of the second passive element **202**. In some embodiments, the second passive element adjacent region **R4** may further include a fifth sub-region **R4e-1** in which the first sub-region **R4a-1** intersects with the third sub-region **R4c**, a sixth sub-region **R4f-1**

in which the first sub-region **R4a-1** intersects with the fourth sub-region **R4d**, a seventh sub-region **R4g-1** in which the third sub-region **R4c** intersects with the second sub-region **R4b-1**, and an eighth sub-region **R4h-1** in which the fourth sub-region **R4d** intersects with the second sub-region **R4b-1**.

The first passive element **201** may be positioned on the second side of the second passive element **202**, and the semiconductor chip **300** may be positioned on the first side of the second passive element **202**. In this case, the roughness of the upper surfaces of the first sub-region **R4a-1** and the second sub-region **R4b-1** of the second passive element adjacent region **R4** may be less than the roughness of the upper surfaces of the third sub-region **R4c** and the fourth sub-region **R4d** of the second passive element adjacent region **R4**. For example, the roughness of upper surfaces of the first sub-region **R4a-1** and the second sub-region **R4b-1** of the second passive element adjacent region **R4** may be less than the roughness of the upper surfaces of the third sub-region **R4c** and the fourth sub-region **R4d** of the second passive element adjacent region **R4** by 80 nm or more.

The roughness of upper surfaces of the fifth sub-region **R4e-1** to the eighth sub-region **R4h-1** of the second passive element adjacent region **R4** may be less than the roughness of the upper surfaces of the third sub-region **R4c** and the fourth sub-region **R4d** of the second passive element adjacent region **R4**. For example, the roughness of upper surfaces of the fifth sub-region **R4e-1** to the eighth sub-region **R4h-1** of the second passive element adjacent region **R4** may be less than the roughness of the upper surfaces of the third sub-region **R4c** and the fourth sub-region **R4d** of the second passive element adjacent region **R4** by 80 nm or more.

By forming the first sub-region **R4a-1**, the fifth sub-region **R4e-1**, and the sixth sub-region **R4f-1** of the second passive element adjacent region **R4** to be relatively flat, the flow of the second flux layer **272** to the first side of the second passive element **202** may be reduced. Accordingly, the second flux layer **272** may be prevented from contaminating the bump pad **124a-1** (refer to FIG. 1A) under the semiconductor chip **300** positioned on the first side of the second passive element **202**.

By forming the second sub-region **R4b-1**, the seventh sub-region **R4g-1**, and the eighth sub-region **R4h-1** of the second passive element adjacent region **R4** to be relatively flat, the flow of the second flux layer **272** to the second side of the second passive element **202** may be reduced. Accordingly, the second flux layer **272** may be prevented from meeting, e.g., contacting, the first flux layer **271**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. 1A) and the package substrate **100**.

FIG. 5 is a plan view showing a semiconductor package **1000d** according to an embodiment. Hereinafter, differences between the semiconductor package **1000c** shown in FIG. 4 and the semiconductor package **1000d** shown in FIG. 5 are described.

Referring to FIG. 5, the semiconductor chip **300** may be positioned on a third side of the first passive element **201**, e.g., the semiconductor chip **300** and the first passive element **201** may be adjacent to each other along the Y direction, and the second passive element **202** may be positioned on a first side of the first passive element **201**, e.g., the first and second passive elements **201** and **202** may be adjacent to each other along the X direction. In other words, the semiconductor chip **300** and the second passive element **202** may be positioned on adjacent (rather than opposite)

sides of the first passive element **201**. In this case, the roughness of upper surfaces of a first sub-region **R2a-1** and a third sub-region **R2c-1** of a first passive element adjacent region **R2** may be less than the roughness of upper surfaces of a second sub-region **R2b** and a fourth sub-region **R2d** of the first passive element adjacent region **R2**. For example, the roughness of upper surfaces of the first sub-region **R2a-1** and the third sub-region **R2c-1** of the first passive element adjacent region **R2** may be less than the roughness of upper surfaces of the second sub-region **R2b** and the fourth sub-region **R2d** of the first passive element adjacent region **R2** by 80 nm or more.

In some embodiments, the roughness of upper surfaces of a fifth sub-region **Re-1** to a seventh sub-region **R2g-1** of the first passive element adjacent region **R2** may be less than the roughness of the upper surface of an eighth sub-region **R2h** of the first passive element adjacent region **R2**. For example, the roughness of upper surfaces of the fifth sub-region **Re-1** to the seventh sub-region **R2g-1** of the first passive element adjacent region **R2** may be less than the roughness of the upper surface of the eighth sub-region **R2h** of the first passive element adjacent region **R2** by 80 nm or more.

By forming the first sub-region **R2a-1**, the fifth sub-region **R2e-1**, and the sixth sub-region **R2f-1** of the first passive element adjacent region **R2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. 1A) and the package substrate **100**.

By forming the third sub-region **R2c-1**, the fifth sub-region **R2e-1**, and the seventh sub-region **R2g-1** of the first passive element adjacent region **R2** to be relatively flat, the first flux layer **271** may be prevented from contaminating the bump pad **124a-1** (refer to FIG. 1A) under the semiconductor chip **300** positioned on the third side of the first passive element **201**.

The first passive element **201** may be located on a second side of the second passive element **202**. In this case, the roughness of the upper surface of the second sub-region **R4b-1** of the second passive element adjacent region **R4** may be less than the roughness of upper surfaces of the first sub-region **R4a**, the third sub-region **R4c**, and the fourth sub-region **R4d**. For example, the roughness of the upper surface of the second sub-region **R4b-1** of the second passive element adjacent region **R4** may be less than the roughness of upper surfaces of the first sub-region **R4a**, the third sub-region **R4c**, and the fourth sub-region **R4d** by about 80 nm or more.

In some embodiments, the roughness of upper surfaces of a seventh sub-region **R4g-1** and an eighth sub-region **R4h-1** of the second passive element adjacent region **R4** may be less than the roughness of upper surfaces of a fifth sub-region **R4e** and a sixth sub-region **R4f** of the second passive element adjacent region **R4**. For example, the roughness of upper surfaces of the seventh sub-region **R4g-1** and the eighth sub-region **R4h-1** of the second passive element adjacent region **R4** may be less than the roughness of upper surfaces of the fifth sub-region **R4e** and the sixth sub-region **R4f** of the second passive element adjacent region **R4** by about 80 nm or more.

By forming the second sub-region **R4b-1**, the seventh sub-region **R4g-1**, and the eighth sub-region **R4h-1** of the second passive element adjacent region **R4** to be relatively

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flat, the flow of the second flux layer 272 to the second side of the second passive element 202 may be reduced. Accordingly, the second flux layer 272 may be prevented from meeting, e.g., contacting, the first flux layer 271. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

FIG. 6 is a plan view showing a semiconductor package 1000e according to an embodiment. Hereinafter, differences between the semiconductor package 1000 shown in FIGS. 1A to 1C and the semiconductor package 1000e shown in FIG. 6 are described.

Referring to FIG. 6, the semiconductor package 1000e may include a first semiconductor chip 301 and a second semiconductor chip 302 on the remaining region R3 of the package substrate 100 (instead of the single semiconductor chip 300 illustrated in FIGS. 1A to 1C). The first semiconductor chip 301 may be located on a first side of a passive element 200, and the second semiconductor chip 302 may be located on a third side of the passive element 200.

In this case, the roughness of upper surfaces of a first sub-region R2a-1 and a third sub-region R2c-1 of the first passive element adjacent region R2 may be less than the roughness of upper surfaces of a second sub-region R2b and a fourth sub-region R2d of the first passive element adjacent region R2. For example, the roughness of upper surfaces of the first sub-region R2a-1 and the third sub-region R2c-1 of the first passive element adjacent region R2 may be less than the roughness of upper surfaces of the second sub-region R2b and the fourth sub-region R2d of the first passive element adjacent region R2 by 80 nm or more.

In some embodiments, the roughness of upper surfaces of a fifth sub-region R2e-1 to a seventh sub-region R2g-1 of the first passive element adjacent region R2 may be less than the roughness of the upper surface of an eighth sub-region R2h of the first passive element adjacent region R2. For example, the roughness of the upper surfaces of the fifth sub-region R2e-1 to the seventh sub-region R2g-1 of the first passive element adjacent region R2 may be less than the roughness of the upper surface of the eighth sub-region R2h of the first passive element adjacent region R2 by 80 nm or more.

By forming the first sub-region R2a-1, the fifth sub-region R2e-1, and the sixth sub-region R2f-1 of the first passive element adjacent region R2 relatively flat, the flow of the first flux layer 271 to a first side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the first semiconductor chip 301 positioned on the first side of the first passive element 201.

By forming the third sub-region R2c-1, the fifth sub-region R2e-1, and the seventh sub-region R2g-1 of the first passive element adjacent region R2 to be relatively flat, the first flux layer 271 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the second semiconductor chip 302 positioned on a third side of the first passive element 201.

FIG. 7 is a plan view showing a semiconductor package 1000f according to an embodiment. Hereinafter, differences between the semiconductor package 1000d shown in FIG. 5 and the semiconductor package 1000f shown in FIG. 7 are described.

Referring to FIG. 7, the semiconductor package 1000f may further include a third passive element 203, a third passive element connecting member 253 connecting the third passive element 203 to the package substrate 100, and

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a third flux layer 273 adjacent to the third passive element 203 on the package substrate 100.

The second passive element 202 may be located on the first side of the first passive element 201, and the third passive element 203 may be located on the third side of the first passive element 201. In this case, the roughness of the upper surfaces of a first sub-region R2a-1 and a third sub-region R2c-1 of the first passive element adjacent region R2 may be less than the roughness of upper surfaces of a second sub-region R2b and a fourth sub-region R2d of the first passive element adjacent region R2. For example, the roughness of upper surfaces of the first sub-region R2a-1 and the third sub-region R2c-1 of the first passive element adjacent region R2 may be less than the roughness of upper surfaces of the second sub-region R2b and the fourth sub-region R2d of the first passive element adjacent region R2 by 80 nm or more.

In some embodiments, the roughness of upper surfaces of a fifth sub-region R2e-1 to a seventh sub-region R2g-1 of the first passive element adjacent region R2 may be less than the roughness of the upper surface of an eighth sub-region R2h of the first passive element adjacent region R2. For example, the roughness of upper surfaces of the fifth sub-region R2e-1 to the seventh sub-region R2g-1 of the first passive element adjacent region R2 may be less than the roughness of the upper surface of the eighth sub-region R2h of the first passive element adjacent region R2 by 80 nm or more.

By forming the first sub-region R2a-1, the fifth sub-region R2e-1, and the sixth sub-region R2f-1 of the first passive element adjacent region R2 relatively flat, the flow of the first flux layer 271 to the first side of the first passive element 201, e.g., along the X direction, may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting, e.g., contacting, the second flux layer 272. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

By forming the third sub-region R2c-1, the fifth sub-region R2e-1, and the seventh sub-region R2g-1 of the first passive element adjacent region R2 to be relatively flat, the flow of the first flux layer 271 to the third side of the first passive element 201, e.g., along the Y direction, may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting, e.g., contacting, the third flux layer 273. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

FIG. 8 is a plan view showing a semiconductor package 1000g according to an embodiment. Hereinafter, differences between the semiconductor package 1000c shown in FIG. 4 and the semiconductor package 1000g shown in FIG. 8 are described.

Referring to FIG. 8, the second passive element 202 may be positioned on the second side of the first passive element 201, and the semiconductor chip 300 may be positioned on the first side of the first passive element 201. In this case, the roughness of upper surfaces of a first sub-region R2a-1 and a second sub-region R2b-1 of a first passive element adjacent region R2 may be less than the roughness of upper surfaces of a third sub-region R2c and a fourth sub-region R2d of the first passive element adjacent region R2. For example, the roughness of upper surfaces of the first sub-region R2a-1 and the second sub-region R2b-1 of the first

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passive element adjacent region R2 may be less than the roughness of upper surfaces of the third sub-region R2c and the fourth sub-region R2d of the first passive element adjacent region R2 by 80 nm or more.

The roughness of upper surfaces of a fifth sub-region R2e-1 to an eighth sub-region R2h-1 of the first passive element adjacent region R2 may be less than the roughness of upper surfaces of a third sub-region R2c and a fourth sub-region R2d of the first passive element adjacent region R2. For example, the roughness of upper surfaces of the fifth sub-region R2e-1 to the eighth sub-region R2h-1 of the first passive element adjacent region R2 may be less than the roughness of upper surfaces of the third sub-region R2c and the fourth sub-region R2d of the first passive element adjacent region R2 by 80 nm or more.

By forming the first sub-region R2a-1, the fifth sub-region R2e-1, and the sixth sub-region R2f-1 of the first passive element adjacent region R2 relatively flat, the flow of the first flux layer 271 to the first side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the semiconductor chip 300 positioned on the first side of the first passive element 201.

By forming the second sub-region R2b-1, the seventh sub-region R2g-1, and the eighth sub-region R2h-1 of the first passive element adjacent region R2 to be relatively flat, the flow of the first flux layer 271 to the second side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting, e.g., contacting, the second flux layer 272. Therefore, an increase in the concentration of the flux mixed with the sealing portion may be prevented (see 400, FIG. 1A). Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented.

FIG. 9 is a plan view showing a semiconductor package 1000h according to an embodiment. Hereinafter, differences between the semiconductor package 1000g shown in FIG. 8 and the semiconductor package 1000h shown in FIG. 9 are described.

Referring to FIG. 9, the semiconductor package 1000h may include the passive element 200, the first semiconductor chip 301, and the second semiconductor chip 302, instead of the first passive element 201, the second passive element 202, and the semiconductor chip 300. The first semiconductor chip 301 may be located on a second side of the passive element 200, and the second semiconductor chip 302 may be located on a first side of the passive element 200. In other words, the passive element 200 may be between the first and second semiconductor chips 301 and 302 along the X direction.

By forming a first sub-region R2a-1, a fifth sub-region R2e-1, and a sixth sub-region R2f-1 of a passive element adjacent region R2 relatively flat, the flow of the flux layer 270 to the first side of the passive element 200 may be reduced. Accordingly, the flux layer 270 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the second semiconductor chip 302 positioned on the first side of the passive element 200.

By forming a second sub-region R2b-1, a seventh sub-region R2g-1, and an eighth sub-region R2h-1 of the passive element adjacent region R2 to be relatively flat, the flow of the flux layer 270 to the second side of the passive element 200 may be reduced. Accordingly, the flux layer 270 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the first semiconductor chip 301 positioned on the second side of the passive element 200.

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FIG. 10 is a plan view showing a semiconductor package 1000i according to an embodiment. Hereinafter, differences between the semiconductor package 1000g shown in FIG. 8 and the semiconductor package 1000i shown in FIG. 10 are described.

Referring to FIG. 10, the semiconductor package 1000i may further include the third passive element 203 on the package substrate 100, the third passive element connecting member 253 connecting the third passive element 203 to the package substrate 100, and the third flux layer 273 adjacent to the third passive element 203 on the package substrate 100. The second passive element 202 may be positioned on the second side of the first passive element 201, and the third passive element 203 may be positioned on the first side of the first passive element 201.

By forming a first sub-region R2a-1, the fifth sub-region R2e-1, and a sixth sub-region R2f-1 of the first passive element adjacent region R2 relatively flat, the flow of the first flux layer 271 to the first side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting, e.g., contacting, the third flux layer 273. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

By forming a second sub-region R2b-1, a seventh sub-region R2g-1, and an eighth sub-region R2h-1 of the first passive element adjacent region R2 to be relatively flat, the flow of the first flux layer 271 to the second side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting, e.g., contacting, the second flux layer 272. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

FIG. 11 is a plan view showing a semiconductor package 1000j according to an embodiment. Hereinafter, differences between the semiconductor package 1000 shown in FIGS. 1A to 1C and the semiconductor package 1000j shown in FIG. 11 are described.

Referring to FIG. 11, the semiconductor package 1000j may include the first semiconductor chip 301, the second semiconductor chip 302, and a third semiconductor chip 303 (instead of the single semiconductor chip 300 in FIGS. 1A-1C). The first semiconductor chip 301 may be located on the second side of a passive element 200, the second semiconductor chip 302 may be located on the first side of the passive element 200, and the third semiconductor chip 303 may be located on the third side of the passive element 200.

In this case, the roughness of top surfaces of a first sub-region R2a-1 to a third sub-region R2c-1 of the passive element adjacent region R2 may be less than the roughness of the upper surface of a fourth sub-region R2d of the passive element adjacent region R2. For example, the roughness of upper surfaces of the first sub-region R2a-1 to the third sub-region R2c-1 of the passive element adjacent region R2 may be less than the roughness of the upper surface of the fourth sub-region R2d of the passive element adjacent region R2 by about 80 nm or more.

The roughness of the upper surfaces of a fifth sub-region R2e-1 to an eighth sub-region R2h-1 of the passive element adjacent region R2 may be less than the roughness of the upper surface of a fourth sub-region R2d of the passive element adjacent region R2. For example, the roughness of

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the upper surfaces of the fifth sub-region R2e-1 to the eighth sub-region R2h-1 of the passive element adjacent region R2 may be less than the roughness of the upper surface of the fourth sub-region R2d of the passive element adjacent region R2 by about 80 nm or more.

By forming the first sub-region R2a-1, the fifth sub-region R2e-1, and the sixth sub-region R2f-1 of the passive element adjacent region R2 relatively flat, the flow of the flux layer 270 to the first side of the passive element 200 may be reduced. Therefore, the flux layer 270 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the second semiconductor chip 302 positioned on the first side of the passive element 200.

By forming the second sub-region R2b-1, the seventh sub-region R2g-1, and the eighth sub-region R2h-1 of the passive element adjacent region R2 to be relatively flat, the flow of the flux layer 270 to the second side of the passive element 200 may be reduced. Accordingly, the flux layer 270 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the first semiconductor chip 301 positioned on the second side of the passive element 200.

By forming the third sub-region R2c-1, the fifth sub-region R2e-1, and the seventh sub-region R2g-1 of the passive element adjacent region R2 to be relatively flat, the flow of the flux layer 270 to the third side of the passive element 200 may be reduced. Accordingly, the flux layer 270 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the third semiconductor chip 303 positioned on the third side of the passive element 200.

FIG. 12 is a plan view showing a semiconductor package 1000k according to an embodiment. Hereinafter, differences between the semiconductor package 1000j shown in FIG. 11 and the semiconductor package 1000k shown in FIG. 12 are described.

Referring to FIG. 12, the semiconductor package 1000k may include the first passive element 201 and the second passive element 202 on the package substrate 100 (instead of the single passive element 200 and the third semiconductor chip 303 in FIG. 11). The semiconductor package 1000k may further include the second passive element connecting member 252 that connects the second passive element 202 to the package substrate 100 and the second flux layer 272 adjacent the second passive element 202 on the package substrate 100.

By forming a first sub-region R2a-1, a fifth sub-region R2e-1, and a sixth sub-region R2f-1 of a first passive element adjacent region R2 relatively flat, the flow of a first flux layer 271 to a first side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from contaminating the bump pads 124a-1 (refer to FIG. 1A) under the second semiconductor chip 302 positioned on the first side of the first passive element 201.

By forming a second sub-region R2b-1, a seventh sub-region R2g-1, and an eighth sub-region R2h-1 of the first passive element adjacent region R2 to be relatively flat, the flow of the first flux layer 271 to a second side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the first semiconductor chip 301 positioned on the second side of the first passive element 201.

By forming a third sub-region R2c-1, the fifth sub-region R2e-1, and a seventh sub-region R2g-1 of the first passive element adjacent region R2 to be relatively flat, the flow of the first flux layer 271 to the third side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting the second flux layer

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272. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

FIG. 13 is a plan view showing a semiconductor package 1000l according to an embodiment. Hereinafter, differences between the semiconductor package 1000k shown in FIG. 12 and the semiconductor package 1000l shown in FIG. 13 are described.

Referring to FIG. 13, the semiconductor package 1000l may include the third passive element 203 and a single semiconductor chip 300 (instead of the first semiconductor chip 301 and the second semiconductor chip 302 in FIG. 12). The semiconductor package 1000l may further include the third passive element connecting member 253 connecting the third passive element 203 to the package substrate 100 and the third flux layer 273 adjacent to the third passive element 203 on the package substrate 100. The second passive element 202 may be located on the second side of the first passive element 201, the third passive element 203 may be located on the first side of the first passive element 201, and the semiconductor chip 300 may be located on a third side of the first passive element 201.

By forming a first sub-region R2a-1, a fifth sub-region R2e-1, and a sixth sub-region R2f-1 of the first passive element adjacent region R2 relatively flat, the flow of the first flux layer 271 to the first side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting the third flux layer 273. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

By forming a second sub-region R2b-1, a seventh sub-region R2g-1, and an eighth sub-region R2h-1 of the first passive element adjacent region R2 to be relatively flat, the flow of the first flux layer 271 to the second side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from meeting the second flux layer 272. Therefore, an increase in the concentration of the flux mixed with the sealing portion 400 (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion 400 (refer to FIG. 1A) and the package substrate 100.

By forming a third sub-region R2c-1, a fifth sub-region R2e-1, and a seventh sub-region R2g-1 of the first passive element adjacent region R2 to be relatively flat, the flow of the first flux layer 271 to the third side of the first passive element 201 may be reduced. Accordingly, the first flux layer 271 may be prevented from contaminating the bump pad 124a-1 (refer to FIG. 1A) under the semiconductor chip 300 positioned on the third side of the first passive element 201.

FIG. 14 is a plan view showing a semiconductor package 1000m according to an embodiment. Hereinafter, differences between the semiconductor package 1000l shown in FIG. 13 and the semiconductor package 1000m shown in FIG. 14 are described.

Referring to FIG. 14, the semiconductor package 1000m may further include a fourth passive element 204 on the package substrate 100, a fourth passive element connecting member 254 connecting the fourth passive element 204 to the package substrate 100, and a fourth flux layer 274 adjacent to the fourth passive element 204 on the package substrate 100. The second passive element 202 may be located on the second side of the first passive element 201,

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the third passive element **203** may be located on the first side of the first passive element **201**, and the fourth passive element **204** may be located on the third side of the second passive element **202**.

By forming a first sub-region **R2a-1**, a fifth sub-region **R2e-1**, and a sixth sub-region **R2f-1** of the first passive element adjacent region **R2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the third flux layer **273**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. 1A) and the package substrate **100**.

By forming a second sub-region **R2b-1**, a seventh sub-region **R2g-1**, and an eighth sub-region **R2h-1** of the first passive element adjacent region **R2** to be relatively flat, the flow of the first flux layer **271** to the second side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. 1A) and the package substrate **100**.

By forming a third sub-region **R2c-1**, a fifth sub-region **R2e-1**, and a seventh sub-region **R2g-1** of the first passive element adjacent region **R2** to be relatively flat, the flow of the first flux layer **271** to a third side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the fourth flux layer **274**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. 1A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. 1A) and the package substrate **100**.

FIG. 15 is a cross-sectional view illustrating a semiconductor package **1000n** according to an embodiment. Hereinafter, differences between the semiconductor packages **1000** shown in FIGS. 1A to 1C and the semiconductor package **1000n** shown in FIG. 15 are described.

Referring to FIG. 15, the semiconductor package **1000n** may include a wire **370** instead of the chip bump **350**. Also, the package substrate **100** may include a wire pad **124a-3** instead of the bump pad **124a-1**. The wire **370** may connect the semiconductor chip **300** to the wire pad **124a-3** of the package substrate **100**.

By forming the remaining region **R3** to be relatively flat, the flow of a flux layer **270** onto the remaining region **R3** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the wire pads **124a-3**.

FIG. 16 is a cross-sectional view illustrating a semiconductor package **1000k** according to an embodiment; Hereinafter, differences between the semiconductor package **1000k** shown in FIGS. 1A to 12 and the semiconductor package **1000k** shown in FIG. 16 are described.

Referring to FIG. 16, the flux layer **270** may not be present on the passive element region **R1** and the passive element adjacent region **R2**. For example, after the passive element **200** is mounted on the package substrate **100**, the flux layer **270** may be removed. For example, when the flux layer **270** includes an aqueous flux, the flux layer **270** may be removed using water.

FIG. 17 is a cross-sectional view illustrating a semiconductor package **1000p** according to an embodiment. Hereinafter, differences between the semiconductor package

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1000 shown in FIGS. 1A to 1C and the semiconductor package **1000p** shown in FIG. 17 are described.

Referring to FIG. 17, the semiconductor package **1000p** may further include a chip flux layer **370-1**. When the semiconductor chip **300** is mounted on the package substrate **100**, the chip flux layer **370-1** around the chip bump **350** is generally used. After the semiconductor chip **300** is mounted on the package substrate **100**, the chip flux layer **370-1** may be removed. For example, when the chip flux layer **370-1** includes an aqueous flux, the chip flux layer **370-1** may be removed using water. However, in another embodiment, as shown in FIG. 17, the chip plus layer **370-1** may remain without being removed.

FIG. 18 is a cross-sectional view illustrating a semiconductor package **1000q** according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIGS. 1A to 1C and the semiconductor package **1000q** shown in FIG. 18 are described.

Referring to FIG. 18, the semiconductor package **1000q** may include the first semiconductor chip **301** and the second semiconductor chip **302** on the package substrate **100** instead of the semiconductor chip **300**. In some embodiments, the first semiconductor chip **301** may be a memory controller chip, and the second semiconductor chip **302** may be a memory chip, e.g., a flash memory chip. The first semiconductor chip **301** may be connected to the bump pads **124a-1** of the package substrate **100** using the chip bumps **350**. The package substrate **100** may further include the wire pads **124a-3** on the fifth insulating layer **121e**. The semiconductor package **1000q** may further include the wire **370** connecting the second semiconductor chip **302** to the wire pad **124a-3** of the package substrate **100**.

Although FIG. 18 illustrates that the semiconductor package **1000q** includes two semiconductor chips, e.g., the first semiconductor chip **301** and the second semiconductor chip **302**, the number of semiconductor chips included in the semiconductor package **1000q** is not limited to two. For example, the semiconductor package **1000q** may include a plurality of second semiconductor chips **302**. The plurality of second semiconductor chips **302** may be a plurality of flash memory chips. The plurality of second semiconductor chips **302** may be stacked on the package substrate **100**. Additionally, or alternatively, the semiconductor package **1000q** may further include a third semiconductor chip, e.g., a DRAM chip, located on, e.g., the package substrate **100**, the first semiconductor chip **301**, or the second semiconductor chip **302**.

By forming the first semiconductor chip **301** and the second semiconductor chip **302** on the remaining region **R3** having a relatively flat upper surface, the flux layer **270** may be prevented from contaminating the bump pad **124a-1** and the wire pad **124a-3**. Further, a flux layer used to mount the first semiconductor chip **301** on the package substrate **100** may be prevented from contaminating the wire pads **124a-3**.

FIG. 19 is a cross-sectional view illustrating a semiconductor package **1000r** according to an embodiment. Hereinafter, differences between the semiconductor package **1000q** shown in FIG. 18 and the semiconductor package **1000r** shown in FIG. 19 are described.

Referring to FIG. 19, the second semiconductor chip **302** may be stacked on the first semiconductor chip **301**. Although FIG. 19 illustrates that the semiconductor package **1000r** includes two semiconductor chips, e.g., the first semiconductor chip **301** and the second semiconductor chip **302**, the number of semiconductor chips included in the semiconductor package **1000r** is not limited to two. For example, the semiconductor package **1000r** may include a

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plurality of second semiconductor chips **302**. The plurality of second semiconductor chips **302** may be a plurality of flash memory chips. The plurality of second semiconductor chips **302** may be stacked on the first semiconductor chip **301**. Additionally, or alternatively, the semiconductor package **1000g** may further include a third semiconductor chip, e.g., a DRAM chip, positioned between the first semiconductor chip **301** and the second semiconductor chip **302** or on the second semiconductor chip **302**.

FIG. **20A** is a cross-sectional view illustrating a semiconductor package **1000s** according to an embodiment. FIG. **20B** is an enlarged view of region **B2** of FIG. **20A**. Hereinafter, differences between the semiconductor package **1000n** shown in FIG. **15** and the semiconductor package **1000s** shown in FIGS. **20A** and **20B** are described.

Referring to FIGS. **20A** and **20B**, the semiconductor package **1000s** may include a wire pad adjacent region **R-1** adjacent to the wire pad **124a-3** and a remaining region **R-2**. The semiconductor chip **300** may be located on the remaining region **R-2**. The roughness of the upper surface of the wire pad adjacent region **R-1** may be less than the roughness of the upper surface of the remaining region **R-2**. Accordingly, the flux layer **270** may be prevented from flowing into the wire pad adjacent region **R-1** and contaminating the wire pad **124a-3**.

FIG. **21** is a flowchart illustrating a method **2000** of manufacturing a semiconductor package according to an embodiment.

Referring to FIGS. **21** and **1A** to **1C**, the package substrate **100** may be roughened in operation **S2100**. Through the operation **S2100**, the roughness of the upper surface of at least one of the first sub-regions **R2a** to **R2d** may be greater than the roughness of the upper surface of a remaining region **R3**.

For example, a mask may be arranged on a region of the package substrate **100** that should be relatively flat, and the region of the package substrate **100** that should be relatively rough may be etched. For etching, dry etching, wet etching, or a combination thereof may be used. For example, plasma etching may be used. For example, a mask may be arranged on the remaining region **R3**, and the passive element region **R1** and the passive element adjacent region **R2** may be etched. The etched passive element region **R1** and passive element adjacent region **R2** may be relatively rough, and a non-etched remaining region **R3** may be relatively flat. Alternatively, a mask may be arranged on the remaining region **R3** and the passive element region **R1**, and the passive element adjacent region **R2** may be etched. As shown in FIG. **2**, the etched passive adjacent region **R2** may be relatively rough, and the non-etched remaining region **R3** and passive element region **R1** may be relatively flat. Alternatively, referring to FIGS. **20A** and **20B**, a mask may be arranged on the wire pad adjacent region **R-1**, and the remaining region **R-2** may be etched. The etched remaining region **R-2** may be relatively rough, and a non-etched wire pad adjacent region **R-1** may be relatively flat.

Additionally, a mask may be arranged on a region of the package substrate **100** that should be relatively rough, and the region of the package substrate **100** that should be relatively flat may be further etched. However, by adjusting the etching conditions, e.g., time, concentration, power, flow rate, pressure, temperature, etc., the etching may be performed weaker than the first etching. For example, a mask may be arranged on the passive element region **R1** and the passive element region **R2**, and the remaining region **R3** may be lightly etched. Alternatively, a mask may be arranged on the passive element region **R2**, and the passive

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element region **R1** and the remaining region **R3** may be lightly etched. Alternatively, a mask may be arranged on the remaining region **R-2**, and the wire pad adjacent region **R-1** may be lightly etched.

Next, the passive element **200** may be mounted on the passive element region **R1** of the package substrate **100** in operation **S2200**. A solder paste may be used to mount the passive element **200** on the passive element region **R1** of the package substrate **100**. The passive element connecting member **250** and the flux layer **270** may be formed of the solder paste. The flux layer **270** may not be removed as shown in FIG. **1A** or may be removed as shown in FIG. **16**.

Next, the semiconductor chip **300** is mounted on the remaining region **R3** of the package substrate **100** in operation **S2300**. Solder balls and flux may be used to mount the semiconductor chip **300** on the remaining region **R3** of the package substrate **100**. The chip bump **350** may be formed of the solder ball. The flux may be removed as shown in FIG. **1A**, or may not be removed as shown in FIG. **17**. As shown in FIG. **15**, alternatively, the semiconductor chip **300** may be connected to the remaining region **R3** of the package substrate **100** using the wire **370**.

Next, the sealing portion **400** covering the package substrate **100**, the passive element **200**, and the semiconductor chip **300** may be formed in operation **S2400**. Finally, the external connection terminal **500** may be attached by attaching a solder ball to the lower surface of the package substrate **100** in operation **S2500**.

By way of summation and review, when a flux layer on a package substrate is mixed with a sealing portion, delamination may occur between the package substrate and the sealing portion. In addition, the flux layer may contaminate a bump pad of the package substrate and a chip bump or a wire pad of the package substrate.

In contrast, embodiments provide a semiconductor package for preventing delamination between a package substrate and a sealing portion and preventing contamination of a bump pad or a wire pad by a flux layer. That is, the roughness of the upper surface of the region adjacent to the passive element may be greater than the roughness of the upper surface of the remaining regions, thereby allowing the flux layer to spread more widely over the relatively rough surface. Accordingly, the concentration of the flux layer mixed in the sealing portion can be reduced and delamination between the package substrate and the sealing portion may be minimized. Also, because the upper surface of the remaining region is relatively flat, it may be difficult for the flux layer to flow into the remaining region, thereby preventing the flux layer from contaminating the bump pad or the wire pad.

Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

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What is claimed is:

1. A semiconductor package, comprising:
 - a package substrate including a passive element region, a peripheral region adjacent to the passive element region, and a remaining region;
 - a first passive element on a non-conductive upper surface of the package substrate in the passive element region;
 - a passive element connecting member connecting the first passive element to the package substrate;
 - a first semiconductor chip on a non-conductive upper surface of the package substrate in the remaining region; and
 - a sealing portion covering the package substrate, the first passive element, and the first semiconductor chip,
 wherein the peripheral region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side of the first passive element, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side of the first passive element,
 - wherein a roughness of a non-conductive upper surface of the package substrate in at least one of the first sub-region to the fourth sub-region is greater than a roughness of the non-conductive upper surface of the package substrate in the remaining region, and
 - wherein the first semiconductor chip overlaps the remaining region in a plan view.
2. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in the passive element region is greater than the roughness of the non-conductive upper surface of the package substrate in the remaining region.
3. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in the passive element region is less than the roughness of the non-conductive upper surface of the package substrate in the at least one of the first sub-region to the fourth sub-region.
4. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in the first sub-region is less than the roughness of the non-conductive upper surface of the package substrate in each of the second sub-region to the fourth sub-region.
5. The semiconductor package as claimed in claim 4, wherein the first semiconductor chip is positioned on the first side of the first passive element.
6. The semiconductor package as claimed in claim 4, further comprising a second passive element positioned on the first side of the first passive element.
7. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in each of the first sub-region and the third sub-region is less than the roughness of the non-conductive upper surface of the package substrate in each of the second sub-region and the fourth sub-region.
8. The semiconductor package as claimed in claim 7, wherein:
 - the first semiconductor chip is positioned on one of the first side and the third side of the first passive element, and
 - a second passive element or a second semiconductor chip is positioned on the other of the first side and the third side of the first passive element.
9. The semiconductor package as claimed in claim 7, further comprising a second passive element and a third

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passive element respectively positioned on the first side and the third side of the first passive element.

10. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in each of the first sub-region and the second sub-region is less than the roughness of the non-conductive upper surface of the package substrate in each of the third sub-region and the fourth sub-region.

11. The semiconductor package as claimed in claim 10, wherein:

- the first semiconductor chip is positioned on one of the first side and the second side of the first passive element, and

- a second passive element or a second semiconductor chip is positioned on the other one of the first side and the second side of the first passive element.

12. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in each of the first sub-region, the second sub-region, and the third sub-region is less than the roughness of the non-conductive upper surface of the package substrate in the fourth sub-region.

13. The semiconductor package as claimed in claim 12, wherein the first semiconductor chip is on one of the first side, the second side, and the third side of the first passive element.

14. The semiconductor package as claimed in claim 13, further comprising:

- a second semiconductor chip and a third semiconductor chip respectively positioned on the other two sides of the first side, the second side, and the third side of the first passive element,

- a second semiconductor chip and a second passive element respectively positioned on the other two sides of the first side, the second side, and the third side of the first passive element, or

- a second passive element and a third passive element respectively positioned on the other two sides of the first side, the second side, and the third side of the first passive element.

15. The semiconductor package as claimed in claim 1, wherein the passive element connecting member is disposed in a recess in the package substrate and contacts sidewalls of the recess.

16. The semiconductor package as claimed in claim 1, further comprising a flux layer disposed on the non-conductive upper surface of the package substrate in at least the second sub-region of the peripheral region and contacting a side surface of the passive element connecting member.

17. A semiconductor package, comprising:

- a package substrate including a wire pad, a peripheral region adjacent to the wire pad, and a remaining region;

- a passive element on a non-conductive upper surface of the package substrate in the remaining region;

- a passive element connecting member connecting the passive element to the package substrate;

- a semiconductor chip on the non-conductive upper surface of the package substrate in the remaining region;

- a wire electrically connecting the semiconductor chip to the wire pad; and

- a sealing portion covering the package substrate, the passive element, the semiconductor chip, and the wire, and contacting a top horizontal surface of the passive element connecting member,

wherein a roughness of a non-conductive upper surface of the package substrate in the peripheral region is less

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than a roughness of the non-conductive upper surface of the substrate in the remaining region, and wherein the semiconductor chip overlaps the remaining region in a plan view.

18. A semiconductor package, comprising: 5
 a package substrate including a passive element region, a peripheral region adjacent to the passive element region, and a remaining region;
 a plurality of external connection terminals on a lower surface of the package substrate; 10
 a passive element on a non-conductive upper surface of the package substrate in the passive element region;
 a passive element connecting member connecting the passive element to the package substrate,
 a memory controller chip on a non-conductive upper surface of the package substrate in the remaining region; 15
 a chip bump connecting the memory controller chip to the package substrate;
 a memory chip on the memory controller chip or the remaining region; 20
 a wire connecting the memory chip to the package substrate;
 a sealing portion covering the package substrate, the passive element, the memory controller chip, the chip bump, the memory chip, and the wire; and 25
 a flux layer disposed between the package substrate and the passive element, the flux layer contacting a side surface of the passive element connecting member, wherein the peripheral region includes a first sub-region on a first side of the passive element, a second sub-

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region on a second side opposite the first side of the passive element, a third sub-region on a third side of the passive element, and a fourth sub-region on a fourth side opposite the third side of the passive element, wherein a roughness of a non-conductive upper surface of the package substrate in at least one of the first sub-region to the fourth sub-region is greater than a roughness of the non-conductive upper surface of the package substrate in the remaining region, wherein the memory controller chip overlaps the non-conductive upper surface of the package substrate in the remaining region in plan view, and wherein the flux layer is disposed on the non-conductive upper surface of the package substrate in the second sub-region of the peripheral region.

19. The semiconductor package as claimed in claim **18**, wherein a width of the at least one of the first sub-region to the fourth sub-region along a direction perpendicular to a thickness direction of the package substrate is about 10 μm to about 500 μm .

20. The semiconductor package as claimed in claim **18**, wherein a difference between the roughness of the non-conductive upper surface of the package substrate in the at least one of the first sub-region to the fourth sub-region and the roughness of the non-conductive upper surface of the package substrate in the remaining regions is 80 nm or more, and

wherein the sealing portion contacts a top horizontal surface of the passive element connecting member.

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